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BOSTON UNIVERSITY

SCHOOL OF MEDICINE

Thesis

ACQUIRING CHEMICAL ATTRIBUTE SIGNATURES FOR GASOLINE: DIFFERENTIATION OF GASOLINE UTILIZING DIRECT ANALYSIS IN REAL TIME – MASS SPECTROMETRY AND CHEMOMETRIC ANALYSIS

by

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B.S., Salve Regina University, 2013
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ACQUIRING CHEMICAL ATTRIBUTE SIGNATURES FOR GASOLINE: DIFFERENTIATION OF GASOLINE UTILIZING DIRECT ANALYSIS IN REAL TIME – MASS SPECTROMETRY AND CHEMOMETRIC ANALYSIS

ASHLEY DAVIS

ABSTRACT

Gasoline is a substance commonly encountered in forensic settings. Unfortunately, gasoline is an easily obtainable ignitable liquid that arsonists commonly use to initiate or expedite the spread of an intentionally set fire. Fires claim the lives of many people each year in addition to causing widespread property damage. Many fire scene investigations result in charges of arson, which has the legal connotation of a committed crime. For this reason, extensive analysis and investigation must be undertaken before any suspected arson scene is deemed an actual case of arson. Although ignitable liquids, including gasoline, may be present at the scene of a fire, it does not necessarily mean they were intentionally used as accelerants. An accelerant is a fuel used to initiate a fire. These realities, in addition to several other factors, demonstrate why a rapid, reliable, gasoline analysis method is crucial to forensic applications. In this thesis, direct analysis in real time – mass spectrometry (DART-MS) is evaluated as a potential method that could better identify, distinguish and classify gasoline brands from one another. Techniques such as DART-MS could enable forensic laboratories to better identify questioned gasoline samples.

Many ignitable liquids share similar chemical properties, and forensically relevant evidence is often obtained from a crime scene in less than favorable conditions. Fire

debris can encompass various materials, including burnt carpet, flooring, items of furniture and clothing, among others. If gasoline was used as an accelerant, it may be present in trace amounts after the termination of the fire. Materials submitted for laboratory analysis may be substrates with compositions that have components similar to those found in some ignitable liquids. These are just a few of the potential obstacles that could be encountered with analyzing fire debris in a forensic setting. Traditionally, gas chromatography – mass spectrometry (GC-MS) methods are utilized for gasoline analysis in the criminal laboratory setting.

While traditional GC-MS methods are sensitive and able to classify samples as gasoline, they are time consuming in terms of both sample preparation and analysis. Additionally, they do not generate differential mass spectral data based on the brand of gasoline. Conversely, gasoline analysis in this research, utilizing the DART-MS method, demonstrated that five different brands of gasoline could be distinguished from one another both by visual examination of mass spectra and with methods of chemometric analysis. Advantageously, the DART-MS method, an ambient ionization technique, requires little sample preparation and a rapid sample analysis time, which could drastically increase the throughput of standard sample analysis with further method development. The goals and objectives of this research were to optimize the DART-MS parameters for gasoline analysis, determine if DART-MS analysis could distinguish gasoline by brand, develop chemometric models to appropriately classify gasoline samples, and finally lay groundwork for future studies that could further develop a more efficient and discriminating DART-MS gasoline analysis method for forensic casework.

Each brand of gasoline was observed to have a chemical attribute signature (CAS) consisting of not only low-mass ions, but also a variety of high-mass ions not usually observed with gasoline samples analyzed by GC-MS. Although variables including season, storage time, dilution and age of the gasoline were observed to contribute to the resulting mass spectral data, once the mass spectra are better understood, they could offer even more discriminating power between samples than simple analysis of the gasoline brand. In this research, DART-MS parameters were first optimized for gasoline analysis. Subsequently, the five acquired brands of gasoline: Shell, Sunoco, Irving, Cumberland Farms and Gulf, were analyzed both undiluted (or neat) and diluted utilizing the DART-MS analysis method. GC-MS data was generated and analyzed to show comparisons. After analyzing the data generated by both approaches, it was apparent that the DART-MS method could generate CASs based on the gasoline brand and offer a degree of differentiation that traditional GC-MS does not.

Additional chemometric analyses utilizing principle component analysis (PCA) and the construction of models with Analyze IQ Lab software verified that the gasoline brands were distinguishable when samples were analyzed with this ambient ionization method. PCA plots of the neat gasoline demonstrated clustering based on brand. Additionally, models constructed from training samples generated from DART-MS analysis of the various brands were able to accurately classify gasoline samples as “yes” or “no” when a test set of gasoline was compared to all five brands. The lowest associated testing error rate for some of these models was 0%. However, additional

analysis with greater sample sizes needs to be further carried out to more accurately evaluate this method of gasoline analysis and classification.

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LIST OF ABBREVIATIONS

° ..... Degrees

°C ..... Degree Celsius

® ..... Registered Trademark

ACS ..... American Chemical Society

amu ..... atomic mass unit

ASTM ..... American Society for Testing and Materials

CA ..... California

CAS ..... Chemical Attribute Signature

CF ..... Cumberland Farms

CDC ..... Centers for Disease Control

cm ..... centimeter

cm^2 ..... centimeter squared

CSV ..... comma separated values

CVS ..... Consumer Value Stores

DART ..... Direct Analysis in Real Time

DART-MS ..... Direct Analysis in Real-Time - Mass Spectrometry

DART-MSD ..... Direct Analysis in Real Time - Mass Selective Detector

EPA ..... Environmental Protection Agency

etOH ..... Ethanol

FBI ..... Federal Bureau of Investigation

FEMA ..... Federal Emergency Management Agency

FL ..... Florida

GC ..... Gas Chromatography

GC-MS ..... Gas Chromatography – Mass Spectrometry

H ..... Hydrogen

H_2O ..... Water

He ..... Helium

HPLC ..... High Performance Liquid Chromatography

IL ..... Illinois

IBM ..... International Business Machines

Inc... ..... Incorporated

LC-MS ..... Liquid Chromatography – Mass Spectrometry

LTQ ..... Linear Trap Quadrupole

M* ..... Metastable

MA ..... Massachusetts

MTBE ..... Methyl Tert-Butyl Ether

MSD ..... Mass Selective Detector

MATLAB ..... Matrix Laboratory

max. inj. time ..... maximum injection time

mL ..... milliliter

min ..... minute

ms ..... millisecond

m/z ..... mass to charge ratio

NY ..... New York

oz. ..... ounce

PTFE ..... polytetrafluoroethylene

PCA ..... Principle Component Analysis

PV ..... polyvinyl

SAV ..... Spectral Attribute Voting

SPME ..... Solid Phase Microextraction

SPSS ..... Statistical Package for the Social Sciences

SVM ..... Support Vector Machine

TAME ..... Tert-Amyl Methyl Ether

TIC ..... Total Ion Chromatogram

™ ..... Trademark

µL ..... microliter

UHPLC-UV ..... Ultra High Performance Liquid Chromatography-Ultra Violet

US ..... United States

USA ..... United States of America

VWR ..... Van Waters and Nat Rogers

XLS ..... eXceL Spreadsheet

1. INTRODUCTION

Fire-related incidents are tragically common both in the United States of America (USA) and worldwide. Many of these occurrences result in investigations to determine if and how a fire was started, if any resulting deaths were caused or intended to be concealed, and if a crime was committed. The setting of intentional fires is a behavior that often leads to involvement of both the legal and mental health systems[1]. According to the Uniform Crime Report of 2013, put forth by the Federal Bureau of Investigation (FBI), 44,840 arson offenses were reported by 15,222 law enforcement agencies, which provided up to one year of data, across the USA [2]. Arson is one of the 8 index crimes used by the FBI to produce the annual crime index, and in the early 1980's, arson experienced a striking upsurge in the USA, surpassing all of the other index crimes in property losses, injuries and deaths[3].

Intentionally set fires are still a concern today, and accelerants, such as gasoline, are frequently used to initiate them. Investigating potential arson cases requires adequate resources as well as a logical and developed method for linking available evidence to a suspect or the appropriate legal offense classification[3]. The objective of this research was to better fulfill this requirement by employing a novel nontraditional method to differentiate gasoline brands utilizing direct analysis in real time – mass spectrometry (DART-MS) in efforts to lay the groundwork for developing a future method better able to link suspect gasoline samples to evidence found at fire scenes. While current methods are both well established and functional, they are also time-consuming, require moderate to extensive sample preparation, and do not produce strongly discriminatory results for

various gasoline samples. A more rapid approach able to generate chemical attribute signatures (CASs) for various gasoline brands could help to better classify questioned samples and increase the analytical throughput.

1.1 Project Goals and Objectives

Fires can lead to devastation resulting in loss of life, property and sense of security. Because gasoline is often used as an accelerant, it has a significant role in fire investigations. The objectives of this thesis include optimizing the analysis of gasoline utilizing DART-MS, determining if different brands of gasoline can be differentiated from one another and developing a faster analysis method for gasoline evaluation that produces CASs for various gasoline samples. Specifically, five brands of gasoline were analyzed both neat and diluted on bare metal mesh and subsequently evaluated by chemometric analysis in efforts to generate models to classify gasoline samples by brand.

Being able to distinguish gasoline brands would offer more discriminatory potential for evidence samples than is available with current methods. Additionally, more rapid analysis of gasoline samples could increase the efficiency and overall throughput of investigations involving fire debris. If DART-MS analysis of gasoline is able to reach these goals, it could enhance gasoline analysis in the laboratory, leading to future improvements in crime laboratory methods, as well as those used in commercial research settings.

1.2 Hypothesis

If DART-MS evaluation allows for differentiation of gasoline brands, then more discriminating evaluation methods could be developed for analyzing gasoline and fire

debris. Faster analysis times utilizing DART-MS could subsequently lead to more efficient gasoline evaluation in crime and research laboratories.

1.3 Forensic Relevance

Arson is a crime with significant impact, causing the lives of many people and animals to be lost every year[4]. From the years 2008 to 2010, the annual average number of deaths was estimated to be 280, while the average number of resulting injuries and the value of property losses were estimated at 775 and $593 million, respectively[4]. These statistics depict the realistic problem of arson and its devastating consequences.

Arson is a legal term with an exact definition that varies from one jurisdiction to another. However, in general, it is a criminal act in which a fire is deliberately set to property, some object or a vehicle[4]. The term "arson" implies criminal and deliberate intent[5]. To avoid a legal connotation, terms such as "intentional fire" or "incendiary fire" are used interchangeably[5]. Due to its legal role in potential criminal prosecutions, the terminology used regarding fire debris analysis has to be carefully chosen[5]. In most jurisdictions, arson is a general intent crime, and by definition, arson does not include any accidentally set fires[1]. Until further investigation of a scene and action by the prosecution that proves intent, the law initially presumes all burning to be accidental[1]. It is important to note that there may be legitimate reasons for the presence of an accelerant at a fire scene[6]. Burning an occupied inhabitance versus an uninhabited dwelling usually incurs more severe punishment[1]. While arson may be suspected, classifying a case as such is avoided until the required burdens have been met to prevent the harsh legal

consequences and connotation of the criminal charge being falsely brought against an individual.

Arson is one of the simplest crimes to commit; neither a weapon nor contact with any other person is necessary[1]. There are often no witnesses, and the crime is commonly committed alone; therefore, criminal intent can be proven with circumstantial rather than direct evidence[1]. The presence of an accelerant is evidence that strongly suggests an intentionally set fire. "Accelerant" is a term that should not be used synonymously with "ignitable liquid."[5] A material is termed an ignitable liquid based on its inherent chemical and physical properties, while an accelerant is defined based on the manner in which it is used[5]. An accelerant is a fuel, which can be and often is a flammable liquid, employed to initiate or propagate the speed or spread of a fire[1]. Consequently, not all flammable liquids that are discovered at a fire scene are accelerants, just as not all accelerants used to commit arson are liquids[5].

Gasoline is one of the accelerants of choice in arson, and gasoline is likely the most commonly encountered accelerant around the world. Gasoline is inexpensive, readily accessible, convenient to transport, easy to use, and in many circumstances, not immediately conspicuous in terms of procurement[5]. Arsonists frequently douse carpets, furniture and other furnishings with gasoline[7]. While other ignitable liquids, such as paint thinners, alcohols, diesel fuels and other solvents, are used to initiate fires, gasoline appears to be the most commonly used accelerant[5]. These facts make gasoline a significant type of evidence. However, gasoline may be present only in minute quantities within fire debris evidence. Ambient ionization mass spectrometry approaches, such as

DART-MS, have been able to detect forensically relevant samples, such as explosives, at nanogram levels[8]. This result shows that high levels of sensitivity are attainable for certain compounds with DART, making it relevant to further forensic evaluation and the method of choice for the gasoline analysis in this research.

As in many other areas of scene investigation, dogs are also utilized in fire incidents. Arson dogs were first employed with the creation of the canine accelerant detection teams in the 1980's by the U.S. Department of Treasury (Bureau of Alcohol, Tobacco and Firearms)[9]. In the future, it may be possible to train dogs to differentiate between brands of gasoline and better identify varying degrees of weathering. While attentive and skilled laboratory analysis of fire debris is necessary to confirm the presence of specific volatile accelerants, arson dogs are useful to alert handlers to trace amounts of ignitable liquids at fire scenes, allowing for locations ideal for sample collection to be identified[9,10]. These trace amounts of ignitable liquids are significant to scene analysis. Often, sections of carpet or flooring that have been burned in a fire are submitted to crime laboratories to determine if an ignitable liquid, possibly an accelerant, is present. Being able to better classify any detected ignitable liquid could help to indicate the source of the fire and could subsequently be linked to evidence samples.

1.4 Gasoline

Ignitable liquids may or may not be employed to initiate or propagate the spread of a fire. The American Society for Testing and Materials (ASTM) has a classification scheme for ignitable liquids including compositional carbon ranges and common m/z peaks observed for each[11]. Frequently used ignitable liquids all possess similar chemical

properties but differ in the boiling point ranges of their components[12]. These differences affect volatility and can cause different components of each ignitable liquid to evaporate at various rates from the sample[12]. Ignitable liquids employed to initiate fires include gasoline, diesel fuels, charcoal lighter fluids, paint thinners and alcohols[5].

Gasoline primarily consists of hydrocarbons, as well as additives and blending agents that are added to improve the performance and stability of gasoline[12,13]. ASTM guidelines note that (fresh) commercial gasoline is a medium-range distillate derived from petroleum typically containing a mixture of hundreds of hydrocarbons ranging from C_4 to C_{12}[47]; as many as thousands of components have been reported to be detected in gasoline[13,14]. Components include various aliphatic (straight-chain) hydrocarbons, cyclopentanes, paraffins, olefins, benzenes and aromatic hydrocarbons including toluene, xylene and a large number of volatile compounds with boiling points <80°C [23,11,14,18,7]. Aromatics comprise a large number of hydrocarbons[13]. Unlike petroleum distillates which have an abundance of n-alkanes, gasoline is differentiated because it has an abundance of aromatics[47]. Over 300 different hydrocarbon molecules used in gasoline were observed to have boiling points ranging from 40°C to 190°C[18], showing that many components are highly volatile. Further classifications of the individual hydrocarbons include the length of the carbon chain backbone, the extent of branching, and the isotopes of carbon atoms[18].

Approximately 2 billion gallons of oxygenates are added to gasoline annually in the U.S., including ethanol (etOH), methyl tert-butyl ether (MTBE), tert-amyl methyl ether (TAME), methanol and other alcohols[15]. However, several states no longer use

MTBE because it was found to contaminate ground water supplies[19]. Additional additives include anti-oxidants, anti-rust agents, metal deactivators, lead scavengers, upper-cylinder lubricants, detergents and dyes; anti-icing additives include short chain n-alcohols, alcohols with long hydrocarbon chains and amines, while detergents include amines and amine carboxylates[14,15]. Lead was previously added to some fuel as an anti-knock agent[15]. Pyrolysis products resulting from high-temperature combustion environments and catalytic events can produce hydrocarbon species not found in the original fuel, such as 1,3-butadiene[15].

Although gasoline is composed of hundreds of components, a wide variety of them are proprietary and differ between different gasoline distributors. Other variations in gasoline can occur due to both seasonal differences in gasoline blends and aging effects. Seasonal differences in gasoline samples have been observed by variations between summer and winter headspace vapor composition[16]. Gasoline vapor pressure is permitted to be relatively high in winter months because when temperatures are colder, evaporation is lower, reducing the emissions of volatile vapors[16]. Specifically, when the n-butane vapor pressure is low and in low abundance in summer gasoline, due to seasonal restrictions on the permissible vapor pressure of anthropogenic volatile organic compounds, it is made exceptionally susceptible to fuel weathering and aging effects, posing additional sources of uncertainty in analysis[16]. The Environmental Protection Agency (EPA) regulates that retailers must begin selling summer-grade gasoline by no later than June 1.

Gasoline also contains nitrogen, sulfur and metals, which introduce molecular instability, even though they are present in low concentrations[21]. This instability causes the oxidation that begins at the production stage, continues throughout storage and transportation, and ultimately results in gum formation[21]. Anti-oxidizers and metal deactivators are added to abate this instability; however, distributors cannot guarantee that gasoline older than approximately two months meets the specifications for engine use[21]. This phenomenon demonstrates that gasoline composition can be affected by aging. This aging effect was further verified by a DART-MS analysis of Coleman camp fuel vapors[39], in which aging was noted to have a considerable effect on mass spectra. Specifically, baseline spectra from "day 1" were compared to spectra obtained from analyzing samples after five months of storage in a sealed container[39]. Studies of aged gasoline also showed that while aromatic compounds were observed to increase (in terms of percent volume) with age, the concentrations of saturated hydrocarbons and benzene were not observed to change significantly[21]. The sample collection and subsequent analysis of fire debris in case-work could be delayed, and this effect should be considered carefully during analysis.

Identifying sources of gasoline is important not only in the interests of forensics and this thesis research, but for purposes related to spills, excess lead violations, contamination problems and boot-legged fuels[22]. For this reason, petroleum dyes, tracers and other additives are present in gasoline and can be used to identify different sources[22]. Dyes used in gasoline include azo-benzene-azo-naphthols, phenyl-azo-napthols (red,

orange, bronze) and anthraquinones (blue) [19]. However, many petroleum specific dyes and their respective structures are proprietary.

1.5 Traditional Gasoline Analysis Methods

The chemical evaluation of fire debris for traces of ignitable liquids can be complex and challenging. Problems arise with accelerant identification, matrix interferences, low contents of target compounds, and preliminary sample manipulation[20]. Several extraction techniques, standardized by the ASTM, are commonly paired with gas chromatography – mass spectrometry and serve as common methods of analysis of fire debris and gasoline in the forensic setting. Extraction techniques include passive and dynamic headspace adsorption, thermal desorption, solvent extraction, and solid phase microextraction (SPME). For the majority of analyses, these methods satisfactorily identify the original accelerant in experimental studies[11,23.24]. Using methods such as these, lower-mass compounds, including toluene, xylenes, C_3 and C_4 alkyl benzenes and naphthenic compounds, are primarily detected[25]. Key disadvantages of SPME analysis include the fragility of the fiber, saturation of the fiber and lack of automation possibilities, among others[26].

1.5.1 Gas Chromatography – Mass Spectrometry

Advances in ignitable liquid analysis have resulted in passive headspace concentration becoming the most commonly used extraction technique for these analyses[27]. Traditionally, in accordance with the ASTM protocol,[23] the method for extracting small quantities of ignitable liquid remnants from fire debris samples is carried

out using activated carbon strips as a suspended sorbent material to collect the residue from the static headspace above the fire debris. Heating from temperatures of 50°C-80°C for 2-24 hours is employed. Longer times and temperatures are required to adsorb compounds with higher boiling points and volatile hydrocarbons present in low quantities. After heated passive headspace concentration, the activated carbon strips are placed into vapor-tight vials and eluted with the desired solvent. Favorable solvents include pentane and carbon disulfide because they do not interfere with the analysis and have higher volatilities than gasoline, allowing for sample concentration by evaporation if needed[11]. Subsequently, the samples are analyzed by GC-MS. The extracted ion profiles are evaluated for retention times and mass spectra characteristic of compound types found in gasoline. The ASTM standard[11] calls for a mass spectrometer capable of acquiring spectra from a mass range of only 40 to 400 amu because the detected ions characteristic of gasoline do not fall in a mass range higher than this[23]; this mass range is much lower than what this research investigated for gasoline CASs generated by DART-MS. The traditional heated passive headspace GC-MS method was performed in this research for comparative purposes.

Passive headspace concentration is advantageous because it is performed in a closed system, avoiding the total consumption of the sample and contamination [28]. A major disadvantage of heated passive headspace concentration is that recovery depends strongly on temperature, time and the nature of the sorbent used, which influences the affinity the sorbent has to the analytes of interest. Since gasoline is composed of many components of various compositions, these factors could prevent some compounds from

being driven into the headspace, as well as the collection of only some components onto the sorbent. Subsequent GC-MS methods can pose additional drawbacks such as the preferential recovery of components, due to variations based on the stationary phase[28]. Often, temperatures greater than approximately 60°C are needed to ensure the recovery of heavier compounds, such as higher boiling-point n-alkanes[28]. Yet, excessive heating and longer sampling times could result in poor yields of the more volatile compounds, due to displacement caused by heavier molecules[28]. While chromatography does separate the components, if for any reason the elution is not completed, carryover can occur[38]. These effects could result in heavier gasoline compounds not being routinely detected, instead producing spectral patterns of lower-mass compounds. Dynamic headspace concentration is another sample collection method that is often followed by GC-MS sample analysis. This method is more rapid but faces similar temperature extraction challenges as those faced by heated passive headspace concentration, such as being subject to breakthrough.

GC-MS analysis still remains a primary analysis method for many forensic samples, including gasoline and other ignitable liquids[30]. While column technology has advanced in recent years, the column stationary phases have different polarities, which could cause some gasoline components, such as additives, to be inadequately separated due to the column properties. A significant problem with GC-MS analysis occurs when target compounds and materials within the background contain ions of the same mass. This problem occurs most frequently when the sample matrices consist of petroleum-based products, especially burnt carpet, causing matrix interferences. These matrices can

produce complex spectra, complicating the comparison of ignitable liquid standards with the ignitable liquid spectra generated from fire debris[30]. Additionally, analysis of an individual sample by GC-MS often takes longer than 10 minutes, not including the sample preparation time[31].

While GC-MS methods are able to differentiate gasoline on the basis of varying degrees of weathering, this differentiation predominantly results from the loss of lighter-mass components as the degree of weathering increases[30]. Chromatographically, petroleum-derived products result in reproducible and recognizable patterns because they are derived from crude oil deposits[27]. Variability in gasoline samples can be attributed to different distributors and refineries supplying to a specific service station and multiple other factors, including location. The composition of gasoline depends on not only the refinery, but also the residual level when the tank was filled[30]. Advances in mass spectrometry have led to the development of novel ambient ionization methods that could be useful in overcoming the challenges met with these traditional methods[20].

1.6 Direct Analysis in Real Time – Mass Spectrometry

DART ionization is a relatively new type of ambient ionization developed by Robert Cody and James Laramee[33]. It has been in commercial use since 2005, but a prototype was proposed as early as 2003[33]. DART is amenable to analyzing solid, liquid and gaseous samples, as well as both polar and nonpolar analytes. It is an ambient ionization source that operates at atmospheric pressure. One novel application of DART is the direct analysis of chemicals on surfaces without the requirement of any sample preparation; compounds are desorbed and subsequently ionized directly from surfaces[33].

Since the initial demonstration of this ionization source concept, DART has successfully analyzed various sample types including chemical warfare agents and their signatures[29], pharmaceuticals[29,33], metabolites[33], peptides[33], synthetic organics[33], drugs of abuse[29, 33], explosives[29, 33], and toxic industrial chemicals[33]. Surface ionization has been performed on: asphalt, human skin, currency, business cards, plastics, body fluids, spices, beverages, glasses and clothing[33]. DART has been used in the rapid analysis of a number of additional materials including soft drinks, pesticides and various edible oils[36,8]. A study of olive oils aimed to distinguish samples of the same substance, encompassing various brands and formulations. This distinction was ultimately accomplished with the edible oils, showing that DART is amenable to differentiating substantially similar compounds[8]. The sampling versatility of this ionization source has proven that it is a powerful atmospheric pressure ion source that has the ability to be combined with mass spectrometry and ion mobility spectrometry[33].

The DART source has a tube segregated into various chambers, which a gas, such as helium, flows through. In the first chamber, an electrical potential is applied to the gas, which produces ions and excited-state species[33]. The gas can also be heated. The DART gas flow can be optionally aimed at and reflected off a sample surface and then subsequently enter the mass spectrometer[23], allowing various options of analysis.

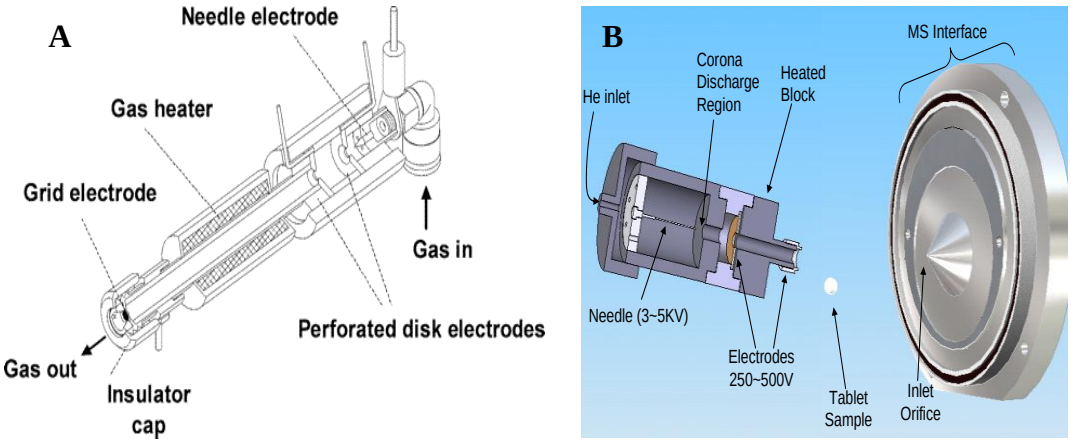

Figure 1. A. Cut away schematic of the DART ionization source, placement of the heater and electrodes are demonstrated[33]; B. Schematic of the DART ionization source including sample placement and inlet into the mass spectrometer (photo courtesy of IonSense, Inc.).

1.6.1 Ionization Method

When the carrier gas enters the source, an electrical discharge produces ions, electrons and excited-state species in a plasma. The excited species consist of metastable (M*) He ions in the excited 2^3S state, which react with atmospheric water to form protonated water clusters that subsequently protonate the analyte[33]; the process is termed Penning ionization[33] and is depicted in Figure 2. Different mechanisms of ionization occur based on the carrier gas, the analyte concentration and the polarity of the ions[33].

Specifically, helium produces $[M+H]^+$ cations, which are usually the primary species detected in the resulting spectra[24].

$$He(2^3S) + H_2O \rightarrow H_2O^{+\bullet} + He(1^1S) + \text{electron}$$

$$H_2O^{+\bullet} + H_2O \rightarrow H_3O^+ + OH^\bullet$$

$$H_3O^+ + nH_2O \rightarrow [(H_2O)nH]^+$$

$$[(H_2O)nH]^+ + M \rightarrow MH^+ + nH_2O$$

Figure 2. Proton transfer reaction and associated equations that occur when helium is used as the carrier gas (equations courtesy of IonSense, Inc.).

The metastable He ions transfer energy only to molecules with lower energy states. If this energy transfer occurs, it results in the formation of a molecular ion and an electron. Because the internal energy of the excited state of the metastable 2^3S helium is 19.8eV, which is higher than the ionization energies of common ambient gases as well as many organic molecules, He is a suitable carrier gas[33].

1.6.2 Hydrocarbon Ionization

DART can successfully ionize a variety of classes of molecules. However, some potential complications exist for the analysis of nonpolar compounds, such as hydrocarbons, which are of crucial interest to this research because they are primary components of gasoline. While DART ionization is effective with polar, basic and acidic compounds, as well as compounds possessing unsaturation sites able to accept a proton, alcohols and saturated hydrocarbons are problematic due to their low proton affinities[35]. Hydrocarbons lack functional groups that are amenable to accepting protons, which could result in an absence of hydrocarbon ion signals[32].

1.6.3 Advantages and Disadvantages

The response of DART is instantaneous, providing real-time results conducive to high-throughput screening methods, and it is more versatile than many other devices. Electronically excited atoms are employed in a few other ion source designs, many of which operate at reduced pressure conditions and expose samples to high electrical potentials that could damage the samples[33]. With DART, samples only need to be exposed to an electrical potential of several kilovolts. DART is amenable to analyzing

various sample types and surfaces, requires little to no sample preparation and is easily interfaced to additional analyzers, such as the mass spectrometer[33]. Progress has also been made to develop a miniature mass spectrometer interfaced to DART, which would be highly beneficial for use at arson scenes because DART can analyze surfaces directly without pretreatment[36]. It is also hypothesized that the surface bombardment by metastable atoms and ionized clusters assists in the desorption process[34].

Disadvantages of DART include its inability to separate sample components in time like many chromatographic methods, and that its accessible mass ranges for some analytes are not known[37,39]. Additionally, the ionization temperature can be crucial in obtaining effective results and if not optimized can result in a poor mass spectral signal[37].

1.7 Analysis of Gasoline by DART-MS

DART serves as a soft ionization technique amenable to a broad range of analytes including polar and non-polar compounds[36]. Various reactions can occur at the point of sample ionization based on the presence or absence of a solvent, the temperature and the electrical potential, which can subsequently affect the ion types and intensities[8,40], these possibilities demonstrate the importance of method optimization prior to sample evaluation. Other studies analyzed undiluted samples prior to dilution to better observe differences in their CASs before the samples were altered[8,40]. This demonstrates the importance of first analyzing neat gasoline samples prior to analyzing any diluted gasoline.

Few other literature sources exist for analyzing ignitable liquids, specifically gasoline, using DART-MS. The few studies analyzing gasoline both carried out analysis

at similar ionization temperatures of 200°C and 250°C[39,40]. Unidentified ions resulting in unique chemical fingerprints were observed in a study of household materials including nylon, drywall and cotton, which were spiked with gasoline and paint thinner and analyzed via DART-MS[40]. Despite their unknown identities, the ions comprising these fingerprints support the differentiation of ignitable liquids using DART-MS methods.

Generally, metastable helium atoms react with atmospheric water to produce protonated water clusters with subsequent proton transfer to the analyte[38], as demonstrated in section *1.6.1 Ionization Method* and Figure 2. However, odd-electron species can also be formed for compounds that do not experience protonation in this manner. Hydride abstraction, or ionization by loss of H^-, results in ions with one less hydrogen atom than other neutral molecules and can occur with gasoline components such as alkanes, which have low proton affinities[39,40]. Such compounds readily fragment forming ion peaks, such as the loss of water, $[(M+H)-H_2O]^+$, in addition to the weak hydride abstraction fragment, $M-H^+$. This phenomenon can create complex mass spectra, containing peaks that result from ions such as various oxidation products[40]. One study proposed that ignitable liquids analyzed by DART-MS are primarily detected by this hydride abstraction process and demonstrated that ions with the form $C_nH_{2n+1}^+$ are derived by hydride abstraction from alkanes with other formulas resulting from the protonation of alkenes and cyclic species[39].

It is important to note that the few studies that also analyzed the CASs of gasoline using DART-MS only generated lower-mass ions, with one study utilizing a scan range

of 50-500 amu and another not reporting on m/z peaks above a mass range of 250 amu[39,40]. These differences between the data from previous studies and the data collected in this research could be attributed to the use of different mass scan ranges and different sample introduction methods. The generation of accurate mass data with DART used in conjunction with high-resolution mass spectrometry for analyzing other compounds has been shown to allow for the identification of the generated ions by determining their molecular formulas in other studies[8,39]. These molecular formulas can be determined because the DART ionization method is soft and frequently results in intact molecules as opposed to fragments[39].

1.8 Chemometric Analysis of Gasoline

Chemometrics extrapolates chemically relevant information from experimental data and utilizes mathematical algorithms to transform data, allowing categorical data to be generated[41]. The goal is then to make accurate classification "predictions" for unknown data[41]. Theory and experience are still integrated into this process because they dictate which variables should be measured as well as how and where to sample[41]. While it is difficult and realistically time restrictive for humans to translate structural variation among a set of molecules to appropriate coordinates that correlate well with the variation of the measured properties, analytical tools can be implemented to allow for the investigation of data in this manner[41]. Chemometric approaches, for example, establish mathematical models relating the experimentally evaluated variables to one another and can then apply those models to explain the variability, noise and patterns in the chemical data inputted[41].

Principle component analysis (PCA) was utilized in this study and is based on multivariate analysis. Multivariate analysis investigates multidimensional data evaluated on a set of similar samples and process points by the "indirect observation of intrinsic 'latent' variables"[41]. PCA modeling can generate unexpected patterns due to the joint effect of all of the variables being taken into account, as opposed to traditional naively observed relationships which only often consider just one or a limited number of variables at one time[41].

Previous studies in gasoline differentiation research have utilized chemometric analysis to differentiate unevaporated gasoline samples analyzed by SPME-GC-MS based on brand. This method was able to detect and differentiate highly diluted gasoline samples based on brand, but the SPME-GC-MS process involved sample preparation, laborious work such as direct injection of the SPME fiber for each sample and a total run time of 70 minutes[22]. One study reported that covariance mapping was able to differentiate fresh gasoline from different sources within a small geographical locale utilizing GC-MS and summed ion mass spectra. However, it was noted that with this analysis method, variations in the chromatographic condition could affect the direct comparison of samples[42].

Analyze IQ Lab software offers the ability to apply chemometric analysis to data obtained from methods such as DART-MS and subsequently construct models to classify data into designated types. Specifically, this program can be utilized to apply various algorithms, such as the proprietary spectral attribute voting (SAV) analysis method, to data to generate the most robust model. The use of chemometric analysis in conjunction

with the rapid DART-MS method can provide a discriminatory capability and the subsequent classification of unknown samples could be developed into a high-throughput method that is more efficient than the traditional time-consuming approaches currently used. Specifically, the SAV analysis method was utilized to build chemometric models of the five gasoline brands analyzed by DART-MS in this research.

2. EXPERIMENTAL DESIGN

2.1 Materials

One-gallon red plastic gasoline containers were purchased from an AutoZone store located at 1160 Broadway Suite B, Sherman Plaza, Saugus, MA 01906. The containers were made in the U.S. and manufactured by the Midwest Can Company in Franklin Park, IL. All of the gasoline was purchased in the state of Massachusetts within the greater Boston area and was of the "87 regular" octane rating. Two sets of each gasoline brand were purchased; the first set was designated as gasoline "set 1," and the second set was designated as gasoline "set 2." The specific brands, addresses and dates of purchase are displayed in Table 1.

Table 1. Brand, source and purchase date information of gasoline samples used for analysis.

Gasoline Brand	Address	Set 1 Purchase Date	Set 2 Purchase Date
Shell	584 Columbus Avenue Boston, MA	November 7, 2014	March 30, 2015
Sunoco	850 Commonwealth Avenue Boston, MA	November 7, 2014	May 18, 2015
Irving	434 Cambridge Street Allston, MA	November 9, 2014	March 30, 2015
Cumberland Farms	148 Chestnut Hill Avenue Brighton, MA	November 9, 2014	March 30, 2015
Gulf	785 Tremont Street Boson, MA	November 7, 2014	May 18, 2015

QuickStrip™ cards were purchased from IonSense, Inc. at 999 Broadway, Suite 404, Saugus, MA 01906 (part number: SVP-7001, revision 1). 4mL glass black-top vials were purchased through Fisher Scientific and manufactured by Kimble™ and Chase™; the associated caps were phenolic with polyvinyl-faced (PV) pulp liners. After experiencing sample loss with these vials, 4mL glass Fisherbrand™ vials with polytetrafluoroethylene (PTFE) lined caps were purchased from Fisher Scientific. The pipette tips used were Fisherbrand™. Plastic and glass transfer pipettes were both used. The 2mL disposable glass Pasteur pipettes were manufactured by PYREX™ USA. Glass 10µL Hamilton syringes were purchased from Fisher Scientific (revision: B, made on October 9, 2014). Glass, 10µL Agilent syringes made in Australia (part number: 9301-0714) were also utilized. 1-5µL Drummond wiretrol disposable micropipettes and the associated dispenser top were used from Drummond Scientific Company, Broomall, PA. Paperclips were purchased from Office Depot and unwaxed dental floss was purchased from CVS. Activated carbon strips (also referred to as activated charcoal strips) were purchased from Albrayco Technologies, Inc. (product designation: ACS-100, lot number: 022114B). Pint-sized (16oz.) aluminum Qorpak™ unlined paint cans with "triple tite lids" were purchased through Fisher Scientific. Silanized GC vials and the associated GC caps and inserts utilized were manufactured by National Scientific. All of the solvents used were from Fisher Scientific. The methanol was UHPLC-UV grade. The pentane used was HPLC grade, and the carbon disulfide was certified ACS from Fisher Scientific. Methylene chloride used was submicron-filtered and in accordance with ACS specifications.

A Corning laboratory stirrer/hot plate (model number: 6795-220, serial number: 480212104) was used for creating evaporated gasoline samples. A VWR Scientific forced air oven (model: 1330FM) was utilized for heated passive headspace concentration. A Vortex Genie-2, manufactured by Scientific Industries, Inc., Bohemia, NY 11716 (model number: G-560, serial number: 2-383096) was employed to vortex all of the samples. A Hirschmann® electric pipetter was used for all of the pipetting carried out with the glass Pasteur pipettes. A rubber mallet and a can opener were utilized for preparing the paint cans associated with the heated passive headspace analysis. Forceps, scissors and razor blades were utilized for cutting and handling any material held directly in front of the DART ionization source by hand.

All of the DART instrumentation was installed and manufactured by IonSense, Inc, 999 Broadway, Suite 404, Saugus, MA 01906 including the following: the DART controller (part number: SVP-100, revision: 1.2, serial number: SVP00436), the DART ionization source (manufactured: August 13, 2014, model: SVP, revision: 5 PO1578, serial number: SVPS00471), the "Vapur by IonSense" interface to the mass spectrometer (part number: SI-140-A-SVP, revision: 2), the linear rail module (rail: 471, serial number: 14314846O1D, revision: 5), the angling module to position the DART ionization source at a 45° angle (part number: JVLC-4509-A, revision: 4) and the tablet carrier module (part number: JVL-2039, revision: 1). The mass spectrometer system utilized was an LTQ (linear trap quadrupole) manufactured by Thermo Fisher Scientific, Inc., 355 River Oaks Parkway, San Jose, CA 95134 (model: Finnigan LTQ Mass Spectrometer System, serial number: LTQ 10158). The gas chromatograph (GC) utilized

for GC-MS was model: Agilent 6890, from Agilent Technologies, Santa Clara, CA. The associated mass spectrometer used with this GC was a 5973 Mass Selective Detector.

DART software, version 4.6.14, was utilized. The LTQ Tune Plus software, version 2.7.0.1094, "LTQ™ 2.7", from Thermo Fisher Scientific was used in conjunction with the LTQ instrumentation. The Xcalibur™ software utilized was "Xcalibur 2.1", Thermo Xcalibur 2.1.0SP1.1160, from Thermo Xcalibur (product version: 1.01.03, release date: July 16, 2009). MSD ChemStation® software, version D.03.00.611, from Agilent Technologies was utilized in conjunction with the Agilent GC. The associated Enhanced Data Analysis ChemStation® software, (version: 61701DA, revision: D.03.00SP1, release date: July 3, 2014) from Agilent Technologies was utilized for data analysis. The software used in association with PCA includes MATLAB® (version R2015a) and IBM SPSS Statistics 22 (product version: 22.0.0.0) designed by IBM Corporation. The "FilterJunk 1.3 LTQ" program, written by IonSense, Inc., was utilized to delete additional page breaks or spaces in data as part of a processing method for input into the chemometric software. An XLS to CSV converter was also employed in the data processing stages in addition to DART-MSD (version: v1.1.0), which nominalized the data. Analyze IQ Lab software, version 2.0, was utilized for chemometric analysis and model building.

2.2 Gasoline Collection and Preparation

Five, one-gallon red, plastic gasoline containers were purchased on the same date from the same AutoZone. Approximately one half-gallon of gasoline was subsequently purchased from each of the following service stations: Shell, Sunoco, Irving, Cumberland

Farms and Gulf at the addresses listed in Table 1. All of the gasoline samples in "set 1" were purchased between November 7, 2014 and November 9, 2014 in the state of Massachusetts. The gasoline samples were stored in their respective gasoline containers.

On November 11, 2014, the first evaporation series was created for each of the five brands. Evaporated samples were created to simulate the various degrees of weathering that gasoline samples can undergo. The evaporation series was readied by first preparing five 4mL aliquots of each gasoline brand into glass vials using plastic transfer pipettes. A different pipette was used for each of the gasoline brands. One of these 4mL aliquot vials was labeled "0%" evaporated for each brand. For each brand, additional clean glass vials were labeled 25%, 50%, 75% and >90%. 1mL at a time was aliquotted into each of the appropriately labeled vials, and a mark was made with a sharpie at the meniscus. This process was performed four times for a total of 4 marked gradations.

The vials were then placed on a hot plate set at a heat setting of approximately 3-3.5 for the 25% and 50% samples. The first gradation corresponds to 0% evaporated. The 25% evaporation samples were taken off of the hot plate when the second gradation from the top of the vial was reached, the 50% evaporation samples at the third gradation, the 75% evaporated samples at the fourth gradation and the >90% evaporated samples at approximately 75% below the fourth gradation mark. After the 25% and 50% samples were removed from the hot plate, the heat setting was raised to approximately 4.5-5 to expedite the evaporation process for the 75% and >90% evaporation samples. Some of the 75% and >90% evaporated samples took longer than one day to evaporate to the

desired evaporation percentage, so not all evaporations were concluded on the same day. The marked gradations and >90% evaporated samples are shown in Figure 3. After the samples were all appropriately evaporated, a dilution series of the 0% evaporated sample was made for each brand in pentane, including 1:100, 1:75, 1:50, 1:25, 1:10 and 1:5 dilutions.

Figure 3. >90% evaporated gasoline samples, in order from left to right: Irving, Gulf, Shell, Cumberland Farms and Sunoco; marked gradations are also shown.

2.3 Optimization of DART-MS Parameters for Gasoline Sample Analysis

The DART-MS parameters that were investigated include the source distance from the linear rail, linear rail speed, ionization temperature, capillary temperature, ionization polarity, scan range, number of microscans and maximum injection time (max. inj. time). In the DART-MS software window, the "QuickStrip" method was selected for all of the optimizations, except for the ionization temperature analysis, or as otherwise noted. The linear rail speed was initially set to 0.5 mm/sec to ensure that the sample did not pass by the ionization source too quickly. The original source distance was set at 2.8 cm from the linear rail. Helium was used as the carrier gas. All of the optimization experiments were performed with gasoline "set 1" unless otherwise stated. The LTQ served as the mass analyzer, and all of the parameters were set within either the Tune™

or Xcalibur™ software programs, with the exception of the linear rail speed and ionization temperature, which were set within the DART software. Unless otherwise noted, both ambient blanks and QuickStrip™ card blanks were collected before any samples were spotted onto the cards and analyzed for all sample analysis and collection throughout this research.

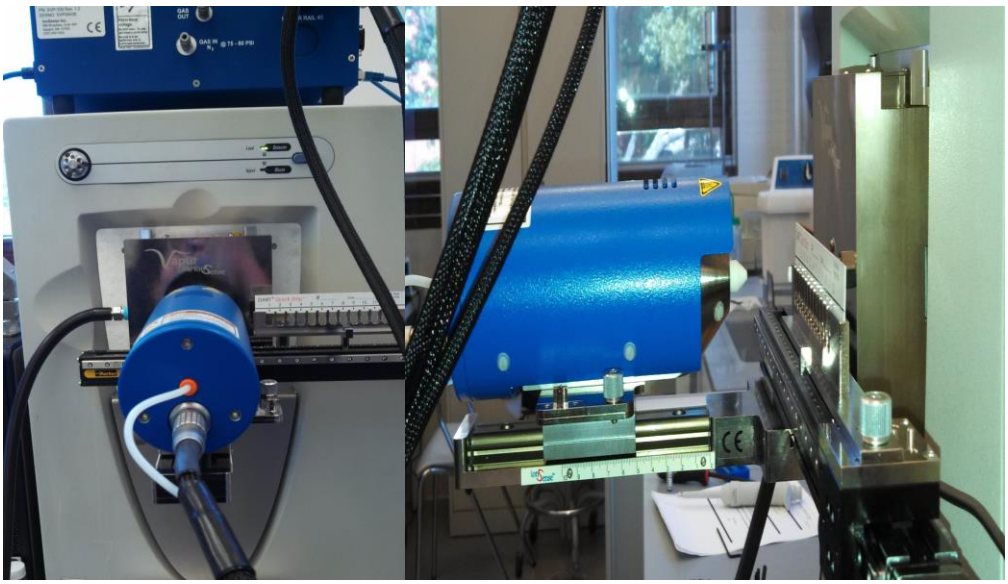

Figure 4. The DART ionization source with associated linear rail module featuring a QuickStrip™ card both head-on (left) and from the side (right).

2.3.1 Ionization Temperature Profile

To determine which temperature produced the optimal mass spectra for the gasoline samples, temperatures were investigated in the range from 150°C to 300°C. Analysis was performed at each 50°C increment. The DART-MS software window was set to utilize the "QuickStrip Temp Profile" instrument setting. The following parameters were utilized in this experiment: 2.8 cm source distance, 0.5 mm/sec linear rail speed,

200°C capillary temperature, positive ion mode, 50-400 atomic mass units (amu) scan range, 1 microscan, 10 ms maximum injection time and centroid peak collection.

As noted previously, ambient background blanks and QuickStrip™ card blanks were collected prior to any sample analysis. All of the brands were analyzed using the 0% evaporated sample at a 1:50 dilution in pentane. A pipette and plastic pipette tips were used to spot 5µL of each sample onto the middle of the bare metal mesh sample window of the QuickStrip™ cards. 250°C was selected as the optimum ionization temperature for future experiments, described in further detail in results sections *3.1 Optimization of DART-MS Parameters for Gasoline Sample Analysis* and *3.2 Gasoline Analysis Using DART-MS*. Additionally, the scan range was increased to include a range of 50-1000 amu to broaden the mass range for observing characteristic ions for the gasoline samples.

2.3.2 Capillary Temperature

After the optimum ionization temperature (250°C) was determined, the capillary temperature was also set to 250°C so that a change in temperature between the ionization source and entrance into the mass analyzer did not occur.

2.3.3 Resolving the Baseline, Microscan Parameters and Reproducibility

After the initial analysis of several background and gasoline sample spectra, subsequent analysis showed an unresolved, noisy baseline. Additionally, not all of the samples were displaying adequate reproducibility. To address these discrepancies, several parameters were investigated: the source distance from the linear rail, the linear

rail speed, the helium gas line settings, the microscan parameters of the ion trap and sample placement on the QuickStrip™ cards. All of these parameters were investigated in a series of analyses. The ambient background and blank QuickStrip™ cards were analyzed under the varied testing parameters first because the aforementioned drawbacks and inconsistencies occurred with these blank samples as well. After determining which parameter needed adjustment based on analyzing the blanks, gasoline samples were subsequently analyzed under the adjusted parameters. The analysis of the gasoline served as a verification that the altered parameter was producing a resolved baseline and reproducibility within the gasoline samples. As noted previously, an ionization temperature of 250°C, a capillary temperature of 250°C and a scan range of 50-1000 amu were utilized from this point forward.

Two helium gas lines were attached to the DART-MS/LTQ instrumentation setup. The collision gas helium line was set at 50 pounds per square inch (psi) and the DART source helium line was set at 75 psi. Many of the preliminary optimization experiments were conducted with both lines activated. The collision gas line was turned off in subsequent analyses because no tandem MS/MS experimentation was performed. However, after turning off this line, a dramatically unresolved baseline in the ambient background blanks, QuickStrip™ card blanks and gasoline samples was observed.

The linear rail speed was set prior to collection of the ambient background in the settings listed in Table 2. This process was then repeated with blank QuickStrip™ cards to determine which parameters resolved the baseline and could produce results that resembled the initial background and sample spectra. Some analyses were repeated to

confirm the reproducibility and confidence in the resultant spectra. After "run 8," details displayed in Table 2, it was observed that having both helium lines activated drastically changed the resulting spectra, particularly at the baseline. This observation was confirmed in "run 9," details again shown in Table 2, in which neat and diluted 0% evaporated Shell samples were analyzed. For this reason, both helium lines were activated for all subsequent analyses to maintain a more resolved baseline and allow for better reproducibility in sample collections.

It was also determined which source distance from the linear rail produced the optimal mass spectra for gasoline, in terms of both a characteristic peak pattern and an optimal ion signal. It was initially hypothesized that the source distance could have been another parameter resulting in the unresolved baseline. The source distances evaluated were 2.8 cm, 3.0 cm, 3.25 cm and 3.5 cm. This optimization was performed in conjunction with the helium gas line optimization described above. 3.25 cm was selected as the optimal source distance.

After determining that the baseline and spectral discrepancies were likely caused predominantly by the helium gas line settings, the linear rail speed was left at the original 0.5 mm/sec setting. This speed was selected to ensure that samples did not pass in front of the ionization source too quickly and were permitted enough time to ionize.

Table 2. Series of parameters investigated to resolve the baseline and optimize DART-MS parameters, includes source distance from linear rail, linear rail speed and He gas line settings.

Run	Parameters
1	2.8 cm, only DART source He on, 0.5 mm/sec rail speed
2	2.8 cm, only DART source He on, 1 mm/sec rail speed
3	3.5 cm, only DART source He on, 0.5 mm/sec rail speed
4	3.0 cm, only DART source He on, 0.5 mm/sec rail speed
5	3.0 cm, only DART source He on, 0.5 mm/sec rail speed
6	2.8 cm, only DART source He on, 0.5 mm/sec rail speed
7	2.8 cm, both DART source and collisional He gas lines on, 0.5 mm/sec rail speed
8	2.8 cm, only DART source He on, 0.5mm/sec rail
9	3.25 cm, both DART source and collisional He valves on, 0.5 mm/sec rail speed*

*Represents an analysis utilizing gasoline samples; the previous runs only utilized blanks

The microscan parameters encompass both the number of scans taken during sample collection as well as the amount of time the ion trap remains open to allow for ion accumulation. The microscan parameters for the ion trap were varied between different data collection sets to determine if changing these parameters produced spectra with higher ion counts. The original parameters employed were 1 microscan and a 10.00 millisecond (ms) maximum injection time. Many of the samples were collected using these parameters for initial visual comparisons of both ion intensity and mass spectral pattern recognitions. However, these parameters were changed to 1 microscan and a 50.00 ms maximum injection time for subsequent sample collection in an effort to evaluate if these changes enhanced the quality of the mass spectra. A third set of parameters, 50 microscans and a 1.00 ms maximum injection time, was also investigated. To prevent future ambiguity between samples due to the varying microscan parameters, experiments were carried out to optimize the microscan parameters by evaluating which

microscan parameters produced the best ion signals and mass spectral patterns for the gasoline samples. To ensure that the gasoline data was reproducible, multiple sample replicates of the various gasoline brands were analyzed at the differing microscan settings outlined above. Additionally, it was questioned whether or not carryover was occurring between samples spotted onto the same QuickStrip™ card. To both assess reproducibility and determine if carryover was occurring between different gasoline brands analyzed on the same QuickStrip™ card, samples were spotted both on individual QuickStrip™ cards and onto QuickStrip™ cards with other brands.

Ambient background blanks and QuickStrip™ card blanks were taken before each sample card was analyzed. All 12 spots of the QuickStrip™ cards were analyzed before samples were spotted to assess any potential contaminants present on the cards. All of the data was collected under the optimized DART-MS parameters, specifically with a 3.25 cm source distance. The microscan parameters varied as previously described.

The first set of data was collected under the following microscan parameters: 50 microscans and a 1.00 ms maximum injection time. One QuickStrip™ card was spotted with both Shell and Sunoco brand gasoline. Specifically, 5µL of each brand at the 0%, 25%, 50% and 75% evaporation percentages was spotted neat onto the QuickStrip™ card. A second QuickStrip™ card was setup in the same manner with Irving and Cumberland Farms brand gasoline. A third QuickStrip™ card was prepared for Gulf brand gasoline using the same methods. Even though Gulf was not placed on a QuickStrip™ card with another brand, it was still spotted for reproducibility purposes.

Plastic pipette tips were used to spot the samples from the original evaporation series created on November 11, 2015 for each brand, from gasoline "set 1".

The microscan parameters for the next set of samples analyzed were 1 microscan and a 50.00 ms maximum injection time. The QuickStrip™ cards were prepared in the same manner as carried out for the prior microscan parameters. However, in this sample set, only the 0%, 25% and 50% evaporation percentages were analyzed because the 75% evaporation percentage took several hours to dry once spotted for each of the brands. In an effort to not allow samples of the other evaporation percentages spotted onto the cards to sit for hours, with the analytes possibly desorbing and evaporating from the card, this evaporation percentage was eliminated.

The final set of data was collected under microscan parameters of 1 microscan and a 10.00 ms maximum injection time. Samples were spotted following the same procedure as the previous microscan analyses, but only one QuickStrip™ card was utilized to spot all of the samples. This deviation was performed due to the determination from the two previous microscan sample sets that carryover was not occurring between brands spotted onto the same card. The 50% evaporated samples were also taking an extended time period to dry, as the 75% evaporation percentage in the previous sample sets was, so only the 0% and 25% evaporation samples of each brand were analyzed in this final microscan parameter sample set.

2.3.4 Elimination of Plastic

To avoid any plasticizers present in the pipette tips or transfer pipettes from interacting with the gasoline samples or complicating the resulting mass spectral patterns,

plastic was eliminated from the experimental methods. Glass Pasteur pipettes replaced the transfer pipettes to transfer gasoline from the plastic containers to the glass vials. Agilent and Hamilton glass syringes replaced pipette tips in the process of spotting the gasoline samples onto the QuickStrip™ cards.

2.4 Gasoline Analysis Using DART-MS

2.4.1 Neat Gasoline Sample Analysis

The optimized parameters determined throughout the previous evaluation studies are displayed in Table 3. All of the data subsequently collected and presented in the "Results and Discussion" were collected under these optimized settings unless otherwise noted. Some modifications that were employed are listed in the following table.

Table 3. Optimized parameters for DART-MS analysis of gasoline.

Parameter	Setting/Value	Special Modifications
Source Distance	3.25 cm	With conical ceramic cap
Source Distance	2.8 cm*	With flat ceramic cap
Linear Rail Speed	0.5 mm/sec	-
Ionization Temperature	250°C	-
Capillary Temperature	250°C	Set to match the ionization temperature
Scan Type	Full scan	-
Scan Range	50-1000 amu	-
Ion Mode	Positive	Negative ion mode utilized for some additional spectra**
Number of Microscans	1 microscan	-
Maximum Injection Time	10.00 ms	50.00 ms for some sample collection sets
Carrier Gas	Helium	Both He lines employed, including collision gas and source gas
Peak Type	Centroid	-

*denotes the parameter was not optimized, changed due to damage of an instrument piece
**denotes data that was collected but not displayed in the results and discussion

All of the undiluted, or neat, gasoline was analyzed using the same basic procedure. After an evaporation series was created for each brand, 5µL samples of each brand at each evaporation percentage were spotted as close to the center as possible in the bare metal mesh QuickStrip™ card window to allow for optimal direct contact with the ionization source. After the conical ceramic cap for the ionization source was broken in an unrelated experiment, a flat ceramic cap replaced it, and the source distance was changed to 2.8 cm for all subsequent analyses. For the original gasoline samples in "set 1", all of the brands were analyzed neat at all of the evaporation percentages. Some samples were collected with microscan parameters of 1 microscan and a 10.00 ms maximum injection time, while some were collected at 1 microscan and a 50.00 ms maximum injection time before the microscan optimization was performed. For "set 2" gasoline samples, all brands were analyzed neat at the 0% and 50% evaporation percentages. All of the samples collected for gasoline "set 2" were collected utilizing glass syringes and microscan parameters of 1 microscan and a 10.00 ms maximum injection time.

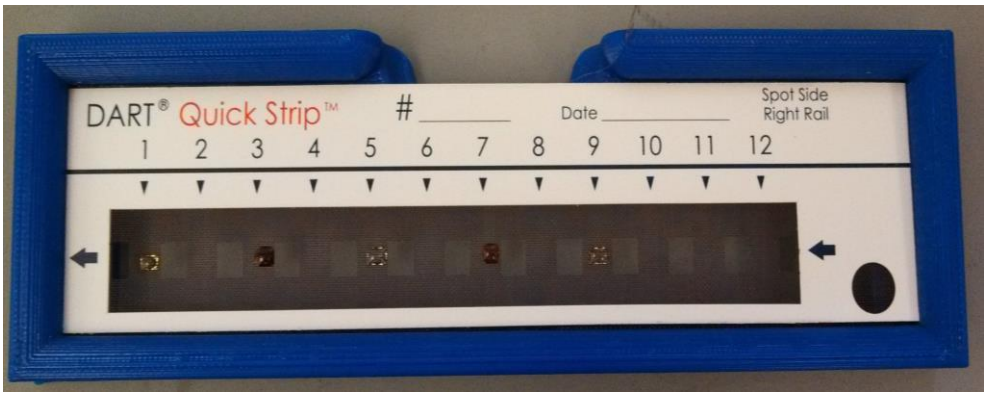

Figure 5. Neat gasoline spotted on a QuickStrip™ card.

The recovery of all of the gasoline brands from "set 1" was also evaluated on the sorbent-coated mesh, Carbopack™ X, and analyzed using the same basic procedure. All of the optimized parameters were used, with a source distance of 3.25 cm and microscan parameters of 1 microscan and a 10.00 ms maximum injection time. Due to the high cost of Carbopack™ X, efforts were made to conserve the sorbent-coated mesh. Because the QuickStrip™ cards have the bare metal mesh set into and bordered by cardboard, the bare metal mesh is conserved and aligned properly with the ionization source. The Carbopack™ X, however, would have required using large strips to align it with the ionization source properly. A one centimeter (cm) wide strip of Carbopack™ X was cut using scissors cleaned with methanol and placed in the linear rail, but as expected, the strip was not positioned high enough to come into contact with the ionization source. Various methods were tested to determine how the Carbopack™ X could be placed in line with the ionization source while conserving material.

To produce more direct contact with the ionization source, the small circular tablet carrier was utilized instead of the linear rail. The one cm wide strip of Carbopack™ X was laid flat in the tablet carrier. Neat and diluted gasoline samples were spotted directly into the center of each tablet spot onto the Carbopack™ X. Both 3µL and 5µL sample aliquots were evaluated. The optimized DART-MS parameters were used, specifically the 3.25 cm source distance and microscan parameters of 1 microscan and a 10.00 ms maximum injection time. The 45° angle module was utilized to position the DART source to hit the sample in a more direct manner.

Additionally, one square centimeter (cm^2) pieces of Carbopack™ X were cut with cleaned scissors. The Shell gasoline "set 1" samples, including neat Shell 0%, Shell 25%, Shell 50%, and Shell 75% evaporated and a 1:50 dilution in pentane of the Shell >90% sample were analyzed in this experiment. 3µL and 5µL aliquots of the samples were spotted over a clean GC vial and allowed to dry. The Carbopack™ X was then held with clean forceps directly in front of the DART ionization source; the "free run" setting was applied in the DART software for this analysis. A Carbopack™ X blank was analyzed first. The forceps were cleaned with methanol in between each sample.

Other methods employed to utilize Carbopack™ X included cutting a one cm wide strip of Carbopack™ X the same length as the QuickStrip™ cards with cleaned scissors. A QuickStrip™ card was deconstructed, and the Carbopack™ X strip replaced the bare metal mesh in the cardboard setting. The Carbopack™ X was subsequently spotted and analyzed in the same manner as the QuickStrip™ cards. One Carbopack™ X strip was spotted with Shell samples at the evaporation percentages: 0%, 25%, 50%, 75% and a 1:50 dilution of >90%. A temperature profile was analyzed for the Shell samples at the 0% and 25% evaporation percentages. The temperature range analyzed was from 200°C to 350°C at 50°C increments. After 250°C was selected as the optimal temperature, Sunoco, Irving, Cumberland Farms and Gulf brand gasoline were analyzed in the same manner as Shell. All of the brands were analyzed at evaporation percentages of 0%, 25%, 50%, and 75%, plus a 1:50 dilution in pentane of >90% evaporated and a 1:5 dilution in pentane of >90% evaporated. For the various analytical procedures

evaluated with Carbopack^{TM} X, the recovery of gasoline analytes was poor, so Carbopack^{TM} X was not utilized in the subsequent analytical procedures.

2.4.2 Analysis of Storage and Age Effects on Gasoline

After all of the parameters were optimized effectively, a new Shell evaporation series was created from gasoline "set 1" on February 19, 2015 in the new, appropriately sealing PTFE lined vials to prevent the unwanted evaporation of gasoline samples that had been diluted. Sample loss by evaporation had been occurring with several of the diluted samples in the PV lined vials, so additional PTFE vials were purchased and subsequently employed. New 1:50, 1:25 and 1:5 dilutions with each of the solvents: pentane, carbon disulfide and methylene chloride were created for the Shell 50% evaporation sample. Prominent m/z 663 and 510 peaks were observed in addition to the characteristic peaks observed in the mass spectra during the initial collection and analysis of the gasoline samples; these characteristic peaks were observed to have dropped to exceedingly low abundances. This trend also was observed in the neat 50% evaporation sample. Subsequently, 50% evaporation samples were created for the other four brands, and the same trend was noticed. Gasoline was then analyzed with a syringe, collecting an aliquot directly from the plastic gasoline container of the Cumberland Farms brand gasoline. The same trend was noticed in the resulting mass spectrum. To address the source of these high-abundance ions and loss of the CAS ions, a new plastic gasoline container was purchased from the same AutoZone as the previous containers. An approximately one cm^2 sample of the container was cut out with a cleaned razor blade

and held with forceps directly in front of the DART ionization source and analyzed under all of the optimized parameters.

After analyzing the plastic container, it was postulated that the gasoline samples had leached compounds out of the plastic container that increased in abundance over time and dominated the resulting mass spectra of the analyzed gasoline samples. It was determined that fresh gasoline samples should be obtained for additional analyses. Three new one-gallon plastic gasoline containers were purchased from the same AutoZone that the original containers were previously purchased from. On March 30, 2015 (date noted to account for any mass spectral changes potentially attributable to seasonal blend), approximately one half-gallon samples each of Shell, Irving and Cumberland Farms brand gasoline were purchased from the same service stations as the first set of gasoline samples ("set 1"). These gasoline samples were designated as gasoline "set 2".

Immediately after purchase, the gasoline was aliquotted into five new glass vials with PTFE lined caps using a different glass Pasteur pipette for each brand. This procedure was performed immediately to prevent any leaching from the plastic container and to avoid potential contamination from the plastic. These vials were intended to serve as gasoline sources if additional evaporation percentages or original gasoline (unevaporated and undiluted) was necessary in future evaluations. New 0% and 50% evaporated samples were created for each of the three brands using the same methods described previously in section *2.2 Gasoline Collection and Preparation*. However, plastic transfer pipettes were replaced with disposable glass Pasteur pipettes, and new glass vials with PTFE lined caps were used. On May 18, 2015 new Sunoco and Gulf

samples were purchased, also considered as part of gasoline "set 2", and the same preparation process was repeated as with the previous three brands.

2.4.3 Solvent Evaluation

For both the 0% and 50% evaporation samples, 1:50, 1:25 and 1:10 dilutions in pentane, carbon disulfide and methylene chloride were created for the Shell, Irving and Cumberland Farms brand gasoline. Pentane was chosen as the best solvent, as further detailed in section *3.2.5 Solvent Analysis*. The 1:50 diluted samples were determined to be too dilute for adequate signal generation and ion pattern recognition, therefore, 1:25 and 1:10 dilutions in pentane were used to collect samples to build models utilizing Analyze IQ Lab software. After the success of the initial models with brands Shell, Irving and Cumberland Farms, 1:25 and 1:10 dilutions in pentane were also created for Sunoco and Gulf for subsequent sample collection and analysis.

2.5 Comparative GC-MS Analysis of Gasoline

All five gasoline brands were analyzed at all five evaporation percentages utilizing a traditional heated passive headspace extraction method followed by a GC-MS method, both in accordance with ASTM guidelines[11,23]. Specifically, heated passive headspace extraction was employed with extraction settings of 80°C for 14 hours followed by GC-MS analysis.

For each extraction, eleven clean aluminum pint-sized cans were labeled appropriately and prepared for analysis, a "blank" or "negative control" can was included for each extraction set. Activated charcoal strips were cut in half using a clean razor blade. The negative control can was assembled first by placing a KimWipe™ into the

bottom of the can so that it took up no more than the bottom 1/3 of the can. Half of an activated charcoal strip was placed onto a slightly opened paper clip, and floss was placed through the paperclip. The floss was then placed over the top of the can, and the can was sealed with a rubber mallet. A clean Kimwipe™ was placed into each of the remaining cans. 25µL of each gasoline sample was spotted onto the Kimwipe™ within each respective sample can. All of the brands at one evaporation percentage were assembled in the same manner as the blank before the next evaporation percentage was prepared.

After the 14-hour extraction, the paint cans were removed from the oven, briefly vented and allowed to cool for several minutes. The activated carbon strips were then desorbed in 1mL of pentane. Each sample was vortexed on the Vortex Genie 2 for approximately 10 seconds on setting 10. A GC-MS sequence was then created using the "ARSON.m" method described below, and pentane was used as a blank between each sample. The parameters of the "ARSON.m" GC-MS method, which was employed for each GC-MS analysis, are displayed in Table 4.

Table 4. ARSON.m method parameters utilized with GC-MS.

Parameter	Setting
Initial Temperature	50°C
Temperature Ramp Up Process	15°C/minute until 280°C was reached (total of 15.33 min) Held at 280°C for 4 min
Inlet Mode	Pulsed Split
Split Ratio	15:1
Split Flow Rate	9.0 mL/min
Pressure	9.92 psi
Total Flow Rate	12.6 mL/min
Average Velocity	31 cm/sec
Solvent Delay	3.00 min
Acquisition Mode	Scan
Low Mass	50.0 amu
High Mass	550.0 amu
Threshold	150

2.6 Chemometric Analysis of Gasoline

2.6.1 Principle Component Analysis

All of the raw spectra data files to be analyzed were exported as comma separated value (CSV) files manually. A MATLAB® program was written by Dr. Pavlovich to assign "0" values for intensity where no peaks existed. The data was then converted to normalized counts and copied into the SPSS software. All of the sample and variable names were changed manually because certain characters, such as capitalized letters, can affect how the program analyzes the data. The appropriate variable settings (such as the components, or dimensions, evaluated) were selected to analyze data in a score plot format based on each individual data set.

2.6.2 Analyze IQ Lab and Model Building

First, a "training" set of data was collected to initially train models based on the types of samples that could be tested against them. All of the samples were collected under the optimized DART-MS analysis parameters, except 2.8 cm was used as the source distance from the linear rail. Xcalibur™ software was utilized to develop a sequence for the collection of each gasoline sample under each variable. For the first three "set 2" gasoline brands (Shell, Irving and Cumberland Farms), 120 samples were collected in the manner shown by Tables 5 and 6. At the beginning of each collection sequence, four ambient blanks were collected. Each QuickStrip™ card was analyzed as a blank so that all 12 spots of each card could be evaluated for the presence of contamination. Once each QuickStrip™ card was evaluated as a blank, it was subsequently spotted in the following manner: spot one served as a card blank with no sample spotted, spot two was spotted with 5µL of pentane to serve as a solvent blank and the following 10 spots were each spotted with 5µL of gasoline samples. The total number of blank samples collected is shown in Table 5. This procedure produced 184 blanks for each brand of gasoline and a total number of 552 blanks for this sample set. The total number of samples collected was 912. Some individual files had to be eliminated due to issues that occurred in the sample window acquisition collection. Therefore, the initial training set was composed of 908 samples, which the original models were trained with and constructed from.

Table 5. Gasoline sample collection details for chemometric analysis.

Evaporation Percentage	Dilution	Number of Samples
0%	1:10	30
	1: 25	30
50%	1:10	30
	1: 25	30

Table 6. Various blanks collected for the chemometric analysis of Shell, Irving and Cumberland Farms gasoline samples.

Blank Type	Number of Samples
Ambient Blank	4
Pentane/Solvent Blank	3
QuickStrip™ Blank	39
TOTAL	**46**

After the initial training set data and testing set data composed of samples from gasoline "set 2" of the Shell, Irving and Cumberland Farms brands generated models with favorable prediction error rates, gasoline "set 2" samples were collected for both Sunoco and Gulf brand gasoline. These samples were collected in the same manner, as outlined by Tables 5 and 6, and discussed in the preceding experimental design methods. This procedure resulted in a training set with a total of 1,488 samples including gasoline samples from all five of the brands and all of the associated blanks.

"Testing" set samples were collected from the same diluted gasoline samples that were created to collect the initial training set, consisting of only the Shell, Irving and Cumberland Farms brands. As in the training set, four ambient blanks were collected at the beginning of the sequence. Only one QuickStrip™ card was run as a blank, and only 10 replicates of each sample type were collected instead of 30, with the omission of the

Shell 0% evaporation at the 1:25 dilution. Table 7 summarizes the samples collected. Each sample card was set up in the same manner as before: the first spot was a card blank, the second spot was a pentane blank and the following ten spots were samples. 5µL of each sample were spotted onto the QuickStrip™ cards and the optimized DART-MS parameters with the 2.8 cm source distance were utilized.

Table 7. Gasoline sample collection details for the Shell, Irving and Cumberland Farms testing set.

Evaporation Percentage	Dilution	Number of Samples
0%	1:10	10
	1: 25*	10
50%	1:10	10
	1: 25	10

*this sample category was not collected for the Shell 0% evaporation due sample loss

All sample data had to be processed into a specific format before being imported into the Analyze IQ Lab software. First, a processing method was added into each of the saved Xcalibur™ sequences that were utilized to collect the various sets of data by adding the processing method column into the sequence and selecting the processing method designed by IonSense, Inc. This processing method integrated and smoothed each peak, allowing the peak to be detected correctly. To apply this processing method to the data, whole sequences were highlighted within the Xcalibur™ software and the "Batch Reprocess" option was selected. The "Filter Junk 3.1 LTQ" program file, created by IonSense Inc., was then applied to each data set by entering in the appropriate file path. This step filtered the data files created from the processing method by removing all unnecessary page breaks and spaces.

The XLS to CSV File Converter program was then employed to convert all sample files into the correct CSV format. After the files were converted into CSV format, they were nominalized using the DART-MSD program; specifically, the "Run Nominalization" action was selected. After all of the files were nominalized, a master folder was created containing all of the nominalized data for each gasoline brand at each of the variables analyzed.

After the training set data was appropriately processed and formatted, it was imported into Analyze IQ Lab. For each model trained with the training data set, a target value list was created that told the model which sample was a "yes" (a positive) and which sample was a "no" (a negative) for each of the models. For example, when any Shell models were constructed, the Shell models had a target value list that designated all Shell samples as "yes" and all blanks, other samples and other gasoline brands as "no". It is important to note that this step is excluded for unknown sample analysis, such as the samples analyzed in the blind study, discussed in section *2.7 Blind Study*. The various algorithms, including linear regression, support vector machine (SVM), k nearest neighbor and the proprietary spectral attribute voting (SAV) algorithm, were applied to the training set data. Each algorithm was trained with the data first with no preprocessing methods. Afterwards, preprocessing methods, including "smoothing", "derivative", "normalize" and "internal standard normalize", were applied in a trial-and-error fashion to the training set data; the default software settings for each preprocessing method were tested first. A cross-validation consisting of five runs and 10 folds was employed on each tested model. With this validation, the software divided the training set data randomly

into 10 sections, or folds. One fold at a time was then used as the testing data and tested against the rest of the training set data. The data was re-randomized into 10 new folds in each run. This process generated the training error rates.

The models with the lowest average training error rates were saved to be tested against the testing data set. The models saved included those generated from the k nearest neighbor algorithm with the "normalize" preprocessing method employed, as well as models generated from the SAV algorithm with the "normalize" preprocessing method employed. The testing data set, outlined in the above methods, was then tested against each of the saved models to evaluate the robustness of each model. The robustness of each model was determined based on the testing error rate, number of false negatives and number of false positives. For each of the gasoline brands, two sets of models were created utilizing the SAV algorithm with the "normalize" preprocessing method. This approach was used because a model was built for each of these brands with only the data from the original training set they were part of, as well as a second model for each brand consisting of all of the gasoline data generated for all of the five brands from all of the training sets. The models producing the best testing error rates were those chosen to use in the blind study for further model validation.

2.7 Blind Study

Fifty samples were prepared for analysis by Dr. Pavlovich and labeled with a coded system to prevent the analyst from knowing the contents of each sample until after the DART-MS analysis and subsequent data input and classification utilizing Analyze IQ Lab software. The unknown samples were analyzed and collected in the same manner as

all of the other samples that were inputted into the Analyze IQ Lab software, with the exception of the use of 5µL glass disposable Drummond pipette tips. Such tips were used to prevent contamination and carryover from unknown sample to sample, as well as to eliminate the possibility of contamination from plasticizers present in pipette tips. All of the unknown samples were analyzed in triplicate.

3. RESULTS AND DISCUSSION

After the five brands of gasoline: Shell, Sunoco, Irving, Cumberland Farms and Gulf were obtained from local service stations in Boston, MA, a series of optimizations was carried out. Subsequently, CASs for the various gasoline sources were analyzed from the resulting mass spectral data generated via DART-MS. Several confounding variables related to the volatility, variability and storage conditions of the gasoline necessitated various stages of evaluation of neat gasoline samples prior to the successive analysis of diluted samples and subsequent construction of classification models utilizing chemometric analysis. Unless otherwise noted, all of the spectra shown here underwent background subtraction to maintain consistency in analysis patterns and to potentially eliminate any background ions with high signals in any of the sample mass spectra.

3.1 Optimization of DART-MS Parameters for Gasoline Sample Analysis

Upon first receiving the gasoline samples, an evaporation series was created for each brand utilizing the methods discussed in section *2.2 Gasoline Collection and Preparation*. A dilution series in pentane was created for each brand including 1:50, 1:25, 1:10 and 1:5 dilutions. Dilutions were created with the objective of detecting CASs for the gasoline brands in dilute solution because sensitivity is crucial in the forensic

setting. Diluted samples were also favorable for the maintenance of the LTQ MS. Two helium lines were connected to the DART and LTQ system. One served as a carrier gas for the DART ionization source, while the other served as a collision gas for the ion trap mass analyzer. With the original background and dilute sample analysis, both helium gas lines were initially activated, and 5µL samples were spotted onto QuickStrip™ cards.

3.1.1 Gasoline DART-MS Temperature Profile

A temperature profile was investigated to determine which DART ionization temperature was optimal for gasoline sample analysis in terms of mass spectral patterns and ion signals. 5µL samples were spotted onto QuickStrip™ cards, both He gas lines were activated and an initial scan range of 50 to 400 amu was employed. Mass spectra were collected for each of the five brands at the following temperatures: 150°C, 200°C, 250°C and 300°C. The 0% evaporation samples at a 1:50 dilution in pentane were utilized for evaluation. The temperature was optimized by evaluating the temperature profiles for each brand; representative mass spectra for the Irving brand are shown in Figure 6.

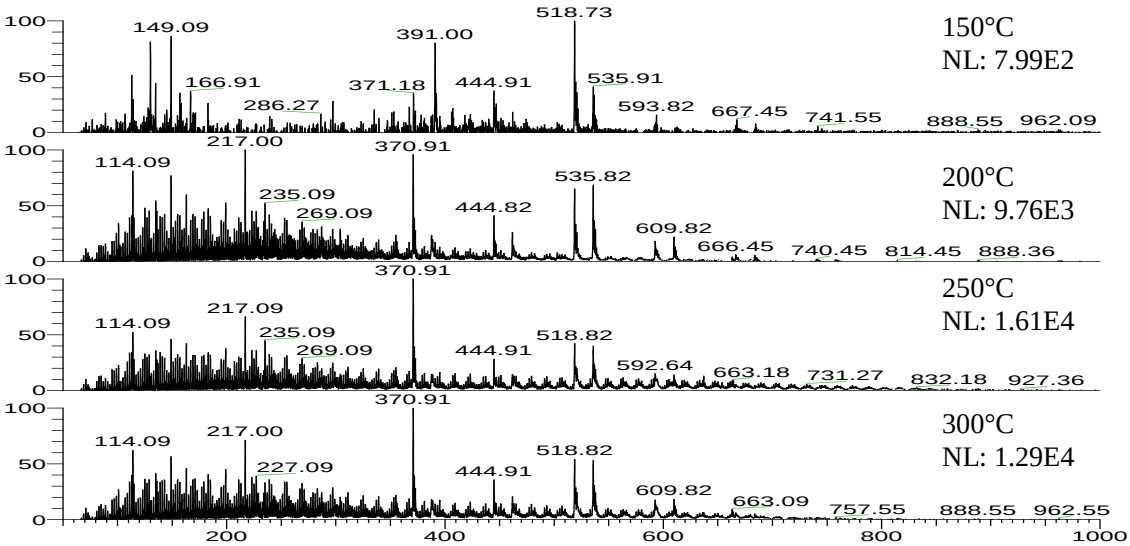

Figure 6. Temperature and associated ion counts, denoted by "NL", for the Irving 0% evaporation in a 1:50 pentane dilution, with pentane blank background subtraction.

While the spectra generated at each of the four ionization temperatures produced the same general ion pattern, the spectra at both the 200°C and 300°C ionization temperature had relatively unresolved baselines. The baseline was not perfectly resolved at the 250°C ionization temperature, but the resolution was better than at the 200°C and 300°C temperatures. While the 150°C ionization temperature generated the most resolved baseline, the signal at this ionization temperature, 7.99 x 10^2 counts (also depicted as NL: 7.99E2 in the spectra formatted by the software, which is the notation displayed in the figures), was the lowest signal observed out of the four temperatures. The signal at the 250°C ionization temperature was found to be the highest, at 1.61 x 10^4 counts, as shown in Figure 6. The optimal temperature was determined to be 250°C based on the factors of both the ion signal and the baseline resolution. Because the ion signal is important to sample analysis, it is a crucial factor to optimize. The temperature profile is displayed only for Irving; however, these conclusions were drawn based on analyzing the temperature profile spectra for all five brands of gasoline.

Analysis at all of the ionization temperatures for the various brands produced low signal counts of only thousands to low tens of thousands. This signal is unfavorable for sample evaluation, and it was postulated that the 1:50 dilution of the gasoline was possibly too dilute to produce proper sample counts and ion pattern generation. Although sensitivity is an important factor, the samples had to be analyzed at concentrations amenable to producing adequate ion counts to allow for initial sample evaluation and pattern analysis. For this reason, more concentrated gasoline samples were subsequently analyzed. Upon optimizing the ionization temperature, the capillary temperature was also adjusted to 250°C to ensure that no analyte experienced a change in temperature between ionization and entrance into the mass analyzer.

3.1.2 Gasoline DART-MS Spectra from Analysis at Varying Helium Conditions

After the ionization temperature was optimized to 250°C and it was determined that 1:50 dilutions may not be concentrated enough for proper preliminary analysis, dilutions of 1:25, 1:10 and 1:5 in pentane were analyzed. The scan range was also increased to 50-1000 amu to evaluate which range produced the most ion-rich characteristic patterns. Additionally, in an effort to conserve He gas, only the He line serving as the DART source carrier gas was activated. Because no tandem MS-MS was being employed, the He line serving as the collision gas was not activated as it was in the previous analysis. Samples were again spotted in 5µL volumes onto QuickStrip™ cards.

The resulting spectra of both gasoline samples and the background were no longer reproducible with initial assessments, and the baseline was dramatically unresolved. A 1:5 dilution in pentane of the 0% Shell evaporation sample is shown below in Figure 7.

This sample demonstrates the highly unresolved baseline and exhibits inconsistencies with the previously generated spectra, as can be seen in Figure 6.

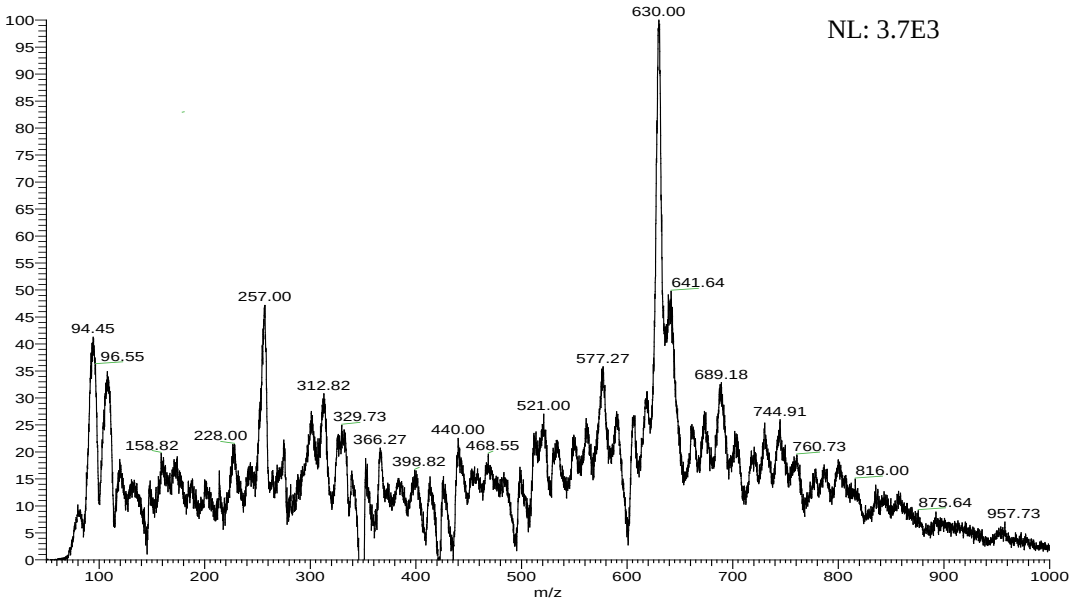

Figure 7. A 1:5 diluted 0% evaporation Shell sample in pentane; the unresolved baseline is noted.

As demonstrated by Figure 7, the sample spectra generated in this manner were not suitable for the additional evaluation of gasoline and for subsequent conclusions to be drawn. To address the source of inconsistencies and the unresolved baseline, several factors, including the source distance from the linear rail, the linear rail speed and the activation of both helium lines were investigated. The source distance could affect the spectra because if the source is too far away from the samples spotted onto the QuickStrip™ card, not all of the analytes may be sufficiently ionized and desorbed. Conversely, if the source distance is too close to the samples spotted onto the QuickStrip™ card, the ionization process and heated carrier gas could cause the ions to disperse and rebound off of the card and linear rail. This effect could result in ions being

directed back towards the ionization source as opposed to the entrance into the capillary and mass analyzer, ultimately resulting in a loss of ions. The linear rail speed could impact the spectra because if the speed is too fast, the samples may not have enough time to be adequately ionized and desorbed. It was conjectured that the baseline discrepancy occurred after the collision He gas line was no longer employed during analysis, based on these observations, the activation of the He gas line serving as the collision gas was also investigated.

To conserve the samples, the background of the QuickStrip™ cards was first analyzed because background spectra were showing the same inconsistencies as the gasoline samples. The observations made with the blanks were then confirmed with the Shell 0% evaporation dilution series as well as the neat Shell 0% evaporation sample. The method parameters were investigated as listed in Table 2 and Figure 8 demonstrates the difference in background spectra when only the DART source helium gas line was activated, as opposed to both He gas lines being activated.

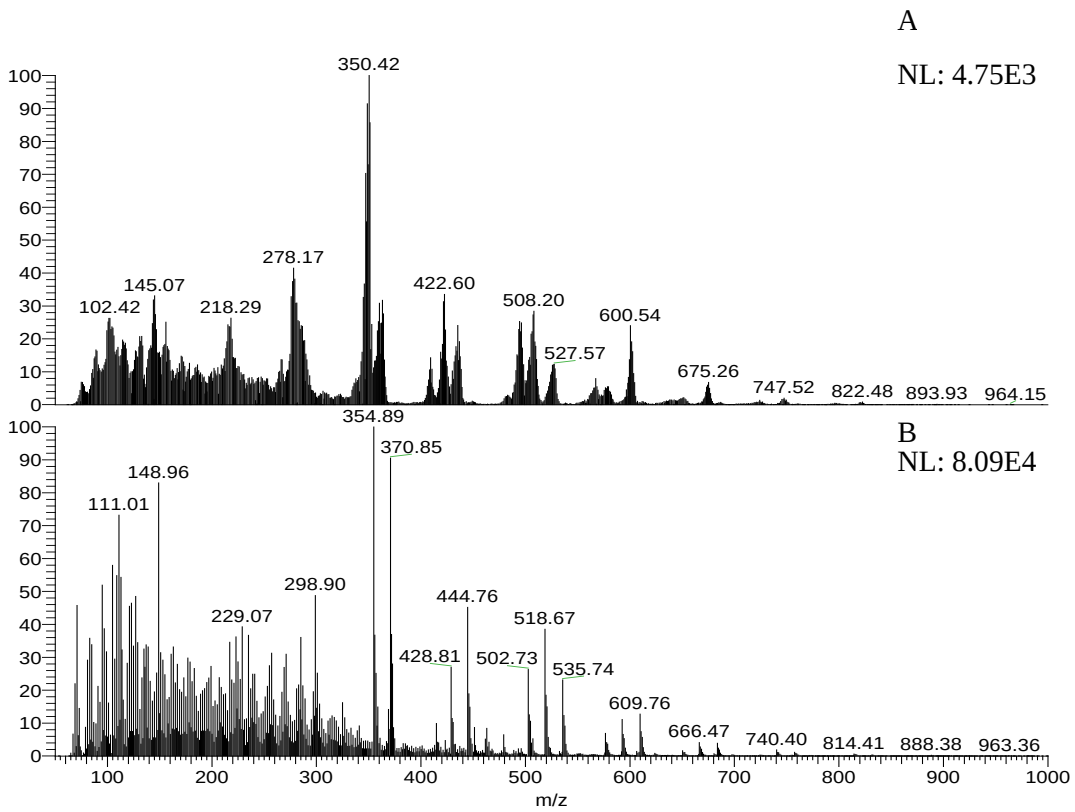

Figure 8. Spectrum A illustrates the observed background with only the DART source He line activated, and B is the observed spectrum when both the DART source He line and collision gas He line are employed.

As demonstrated by Figure 8, both He gas lines being activated appeared to produce the greatest spectral change and addressed the issue of resolving the baseline. Quadrupole ion trap mass spectrometry has the advantage of a background gas pressure that is higher to some extent than what is typically used for mass spectrometry[49]. Helium bath gas is needed for ion trap mass analysis to improve resolution[50], as well as sensitivity[49], and is already present within the system at a static background pressure even without induced collisions[50]. For this reason, the collision gas used is typically Helium[50]. The bath gas assists the capture of ions that are injected into the oscillating quadrupole field, subsequently cooling the ions into the center of the ion trap prior to mass-selective injection[51]. While the trapping process of the ion trap is not well established, the

presence of bath gas in the ion trap produces significant increases with regard to trapping efficiencies[49]. The collision gas line, therefore, also serves as the bath gas in the quadrupole linear ion trap configuration that was utilized in this study and offers an explanation as to why the baseline appeared unresolved when the He collision gas line was not employed in this research.

3.1.3 Gasoline DART-MS Spectra from Analysis at Various Source Distances

As previously noted, it was hypothesized that the source distance could also cause the spectral discrepancies. After analyzing the various source distances listed in section *2.3.3 Resolving the Baseline, Microscan Parameters and Reproducibility*, in addition to the linear rail speed and the He lines setup, the best source distance determined from analyzing the QuickStrip™ cards was then utilized for gasoline sample analysis. Both He lines were activated and the optimized linear rail speed was utilized for this gasoline sample analysis; the resulting spectrum is shown in Figure 9.

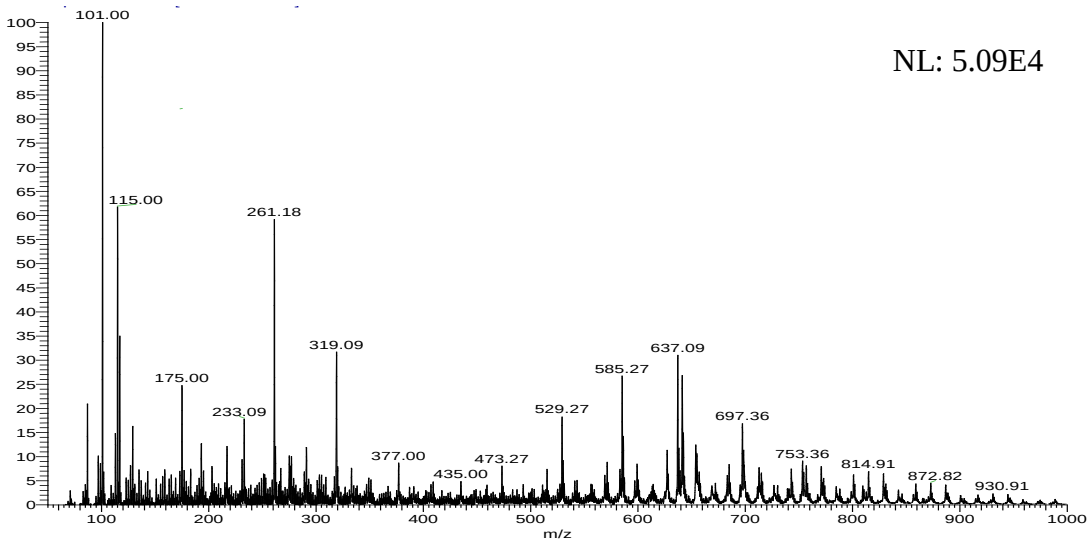

Figure 9. Shell 0% evaporation neat gasoline sample spectrum analyzed at a 3.25 cm source distance, a 0.5 mm/sec linear speed and activation of both He gas lines.

As shown in Figure 9, the spectral pattern of gasoline is optimal at a source distance of 3.25 cm and a linear rail speed of 0.5 mm/sec; adequate counts were also achieved at 5.09 x 10^4, showing an improvement in the ion count with the neat gasoline, as opposed to the lower counts observed in Figure 6 with the 1:50 diluted samples. This spectrum also demonstrated that the He gas line observation was verified with a gasoline sample. Both He gas lines were activated for all subsequent analyses and were included in the optimized parameters, in addition to a source distance of 3.25 cm and a linear rail speed of 0.5 mm/sec. Other studies[9] also postulated that the sample placement and quantity, or volume, can affect the relative abundances of both the molecular ion and the protonated molecules. The study proposed that this effect may arise because a sample placed further away from the DART source may form a considerable number of protonated water clusters; however, the survival of metastable excited helium atoms is improbable[9]. This situation would result in fewer protonated analyte species, causing a decrease in the signal of the resultant ion patterns[9].

3.1.4 Gasoline DART-MS Spectra at Varying Microscan Parameters

Quadrupole ion traps are instruments that function as ion storage units, which detain gaseous ions for a period of time, and as mass spectrometers, in which the m/z ratio of the confined ions can be measured[43]. The mass spectrum is derived from averaging the spectrum measured during each microscan acquired during sample analysis[44]. Specifically, the "number of microscans" parameter represents the number of times that the ions in the ion trap are scanned; subsequent averaging of these scans produces the resultant mass spectrum[44]. Decreasing the number of microscans that are

averaged to generate a mass spectrum increases the scan rate at the cost of a decreased signal-to-noise ratio and mass spectral reproducibility[45]. The maximum injection time is the maximum amount of time that ions are permitted to accumulate in the ion trap[46].

While the original microscan parameters were set to obtain 1 microscan with a maximum injection time of 10 ms for sample collection, it was hypothesized that increasing the maximum injection time could enhance the quality of the spectral results by permitting more ions into the trap to increase signal and generate ion-rich spectra, particularly with the diluted samples. However, reducing the maximum injection time increases the sensitivity by reducing the number of ions in the trap, as demonstrated in literature[43,46]. For the preliminary gasoline analysis, sensitivity was not a concern because the objective was to first determine if CASs could be generated for each brand of gasoline. Moving forward in future development, however, sensitivity will play a significant role in the forensic science applications of this approach because evidence samples are usually not of ideal quality or quantity, and gasoline residues could be present in low concentrations.

To determine if increasing the maximum injection time improved the gasoline sample mass spectra, the scan parameters were later amended to 1 microscan and a maximum injection time of 50 ms to see if a longer maximum injection time would produce better ion signals and patterns as previously hypothesized. Some gasoline samples were collected and evaluated using these parameters. To ensure reproducibility and consistency in all of the samples utilized for chemometric analysis, the microscan

parameters were optimized prior to any data collection for the Analyze IQ Lab software and model construction.

A series of experiments was carried out with a new evaporation series, created from "set 1" Shell gasoline, at the varying microscan parameters listed in the experimental design section, *2.3.3 Resolving the Baseline, Microscan Parameters and Reproducibility*, including an additional parameter investigated with a collection of 50 microscans with a 1.0 ms maximum injection time. Analyses were carried out for all of the gasoline brands, with representative results for the Shell gasoline samples shown in Figure 10.

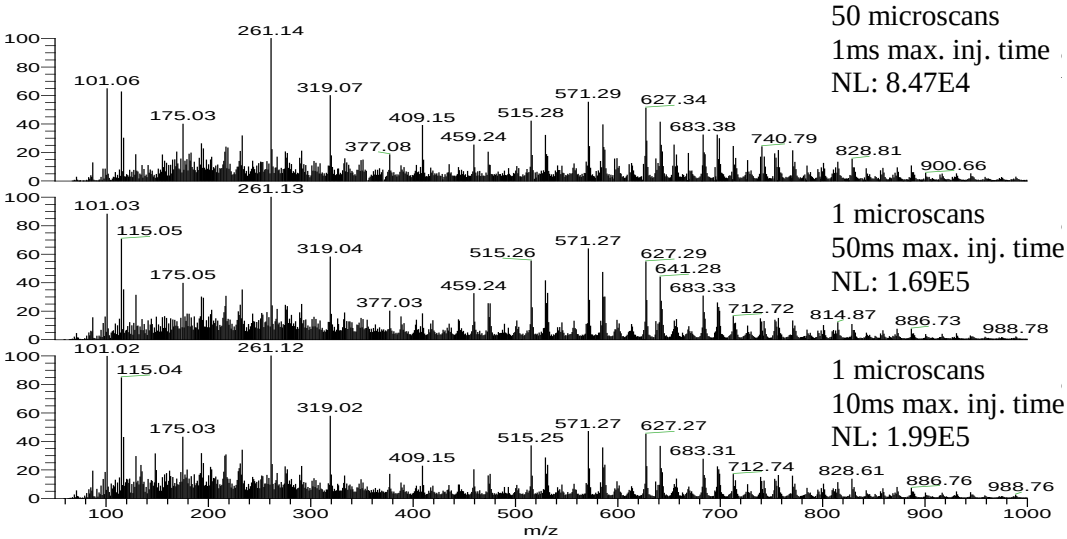

Figure 10. Shell 25% evaporation, neat, analyzed at three different sets of microscan parameters, associated ion counts displayed.

As Figure 10 shows, the overall general CAS was similar for the neat Shell 25% evaporation sample spectra derived from all three sets of microscan parameters. This result demonstrates that the CAS is reproducible for the Shell gasoline despite variations in the microscan parameters. The optimized parameters were therefore determined based

on the ion count signal. The spectra exhibited ion count signals of 8.47×10^4, 1.69×10^5 and 1.99×10^5, as shown from top to bottom in Figure 10, respectively. The original microscan parameters of 1 microscan and a 10.00 ms maximum injection time were determined to be the optimal microscan parameter conditions because these produced the highest ion signals. These parameters were utilized for all of the subsequent sample collection and analysis, including collecting samples for all of the models built during chemometric analysis.

3.1.5 Reproducibility and Carryover Concerns

In conjunction with altering the microscan parameters, the reproducibility in sample collection and analysis was investigated. Several evaporation percentages of all of the gasoline brands were analyzed both on individual QuickStrip™ cards, spotted only with other samples of the same brand, and QuickStrip™ cards spotted with samples from multiple brands, as outlined in section *2.3.3 Resolving the Baseline, Microscan Parameters and Reproducibility*. This experimental design was employed to determine the reproducibility on a sample-to-sample basis and also to evaluate whether more than one gasoline brand being present on a QuickStrip™ card produced any carryover to other brands that could result in altered mass spectra. This "individual versus same-card" analysis was carried out with analysis at both 50 microscans and a 1 ms maximum injection time, and with 1 microscan and a 50 ms maximum injection time, to additionally confirm the sample reproducibility and verify that carryover between brands was not occurring under any conditions. Some of the resulting spectra from this optimization process are displayed in Figure 11.

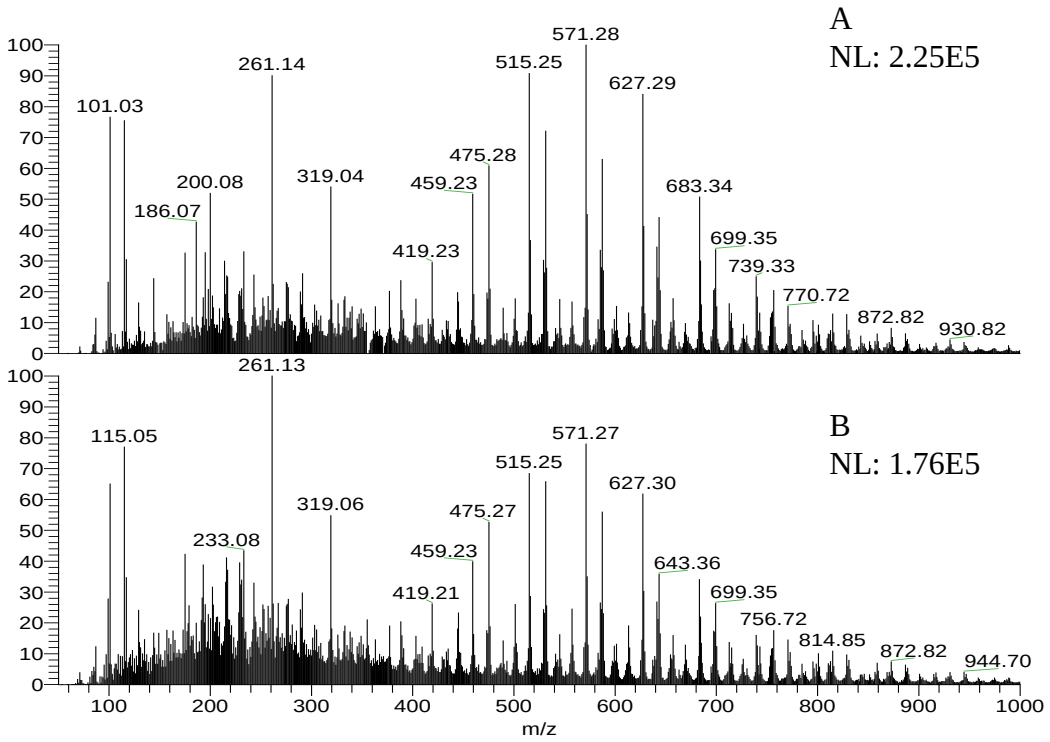

Figure 11. Shell 50% evaporation samples, microscan parameters of 1 microscan with a 50 ms max. inj. time; spectrum A shows a Shell sample spotted on an individual card, spectrum B shows a Shell sample spotted on the same card as Sunoco samples.

Although these evaluations were carried out for multiple brands of gasoline at various evaporation percentages and microscan parameters, all of the sample analysis suggested the same result: no carryover was occurring between brands, and spotting different brands on the same card was confirmed to have no significant effect on the resultant mas spectra of the other brands present on the card. As the Shell sample spectra in Figure 11 show, the reproducibility of the CAS obtained for each gasoline sample was also validated.

After optimizing all of the parameters and addressing concerns with reproducibility and resolving the baseline, neat gasoline samples were analyzed under the optimized parameters to determine the CASs of each brand and assess the level of

differentiation that could be achieved visually by MS data generated with the novel application of the DART ionization method. Because the signal and uniqueness of the patterns derived with the diluted gasoline samples were questioned, the CASs for different brands were initially assessed by evaluating the neat gasoline samples.

These optimization procedures illustrate the importance of sample optimization with DART-MS before definitive conclusions can be deduced regarding an analyte. Several factors can explain why a low ion signal could occur or why the expected ions could be absent.

3.2 Gasoline Analysis Utilizing DART-MS

Due to the relatively poor ionization of hydrocarbons via DART-MS, the proprietary nature of some of the specific gasoline formulations and additives, and the limited previous literature related to both the generation and identification of higher-mass ions in gasoline, the initial analysis consisted of acquiring and observing the overall mass spectral patterns between the gasoline brands in a qualitative manner. These initial evaluations were performed to determine if discriminatory pattern spectra were produced between the varying brands of gasoline and determine how they could be utilized and potentially identified for future applications in fire debris analysis and method development.

3.2.1 Neat Gasoline (Set 1) On QuickStrip™ Cards Analyzed via DART-MS

CASs were observed and evaluated with the original gasoline, designated “set 1.” These samples were first analyzed neat to establish the inherent CASs of each of the five brands. Diluted sample analysis was postponed to ensure that the ion-rich mass spectrum

of each brand was observed in full, and to obtain favorable ion count signals. Once the characteristic patterns were observed and reproduced, diluted samples were later analyzed.

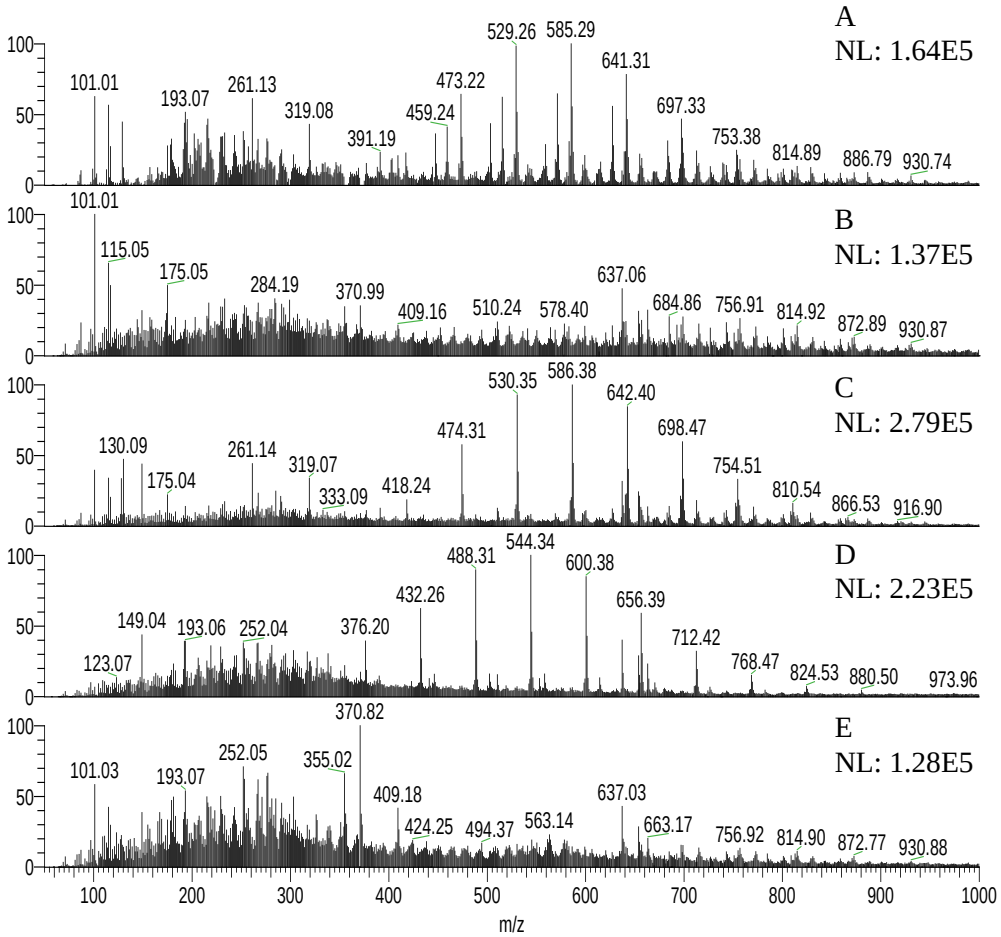

Figure 12. All of the "set 1" gasoline brands analyzed at the optimized parameters, neat, at the 50% evaporation; spectra A corresponds to Shell, B to Sunoco, C to Irving, D to Cumberland Farms and E to Gulf.

Although each brand was evaluated at 0%, 25%, 50%, 75% and >90% evaporations, the 50% evaporation samples proved to be the most different between the brands. This result likely occurs because many of the lighter hydrocarbon components present in all of the gasoline samples, such as toluene and xylenes, were likely lost during

the evaporation process, leaving the higher-mass additives that differ between the brands remaining. However, not as many compounds are lost with the 50% evaporation as with the 75% and >90% evaporations. This hypothesis explains why the 50% evaporation offers a greater number of differentiating compounds between the gasoline brands. The more compounds present in the sample, the more discriminatory power the mass spectra can offer for comparison between brands. It was also observed that each brand of gasoline had a characteristic "bell-curve" pattern consisting of ions in the higher mass range, spanning from m/z values in the mid-400's to the high-900s, as shown in Figure 12. The bell-curves of Sunoco and Gulf are less prominent and are spectrally more similar to one another than those of the other brands.

Some of the lighter and more volatile compounds are lost in the more evaporated samples, similar to the GC-MS analysis of more weathered samples where the earliest-eluting peaks, representative of compounds such as xylenes, are no longer present. However, it was expected that there would be an overall trend within each gasoline brand across all evaporation percentages, especially with regard to the higher-mass compounds, which, as noted previously, are likely additives specific to each gasoline brand. Figure 13 shows Shell ("set 1") brand gasoline at all five evaporation percentages.

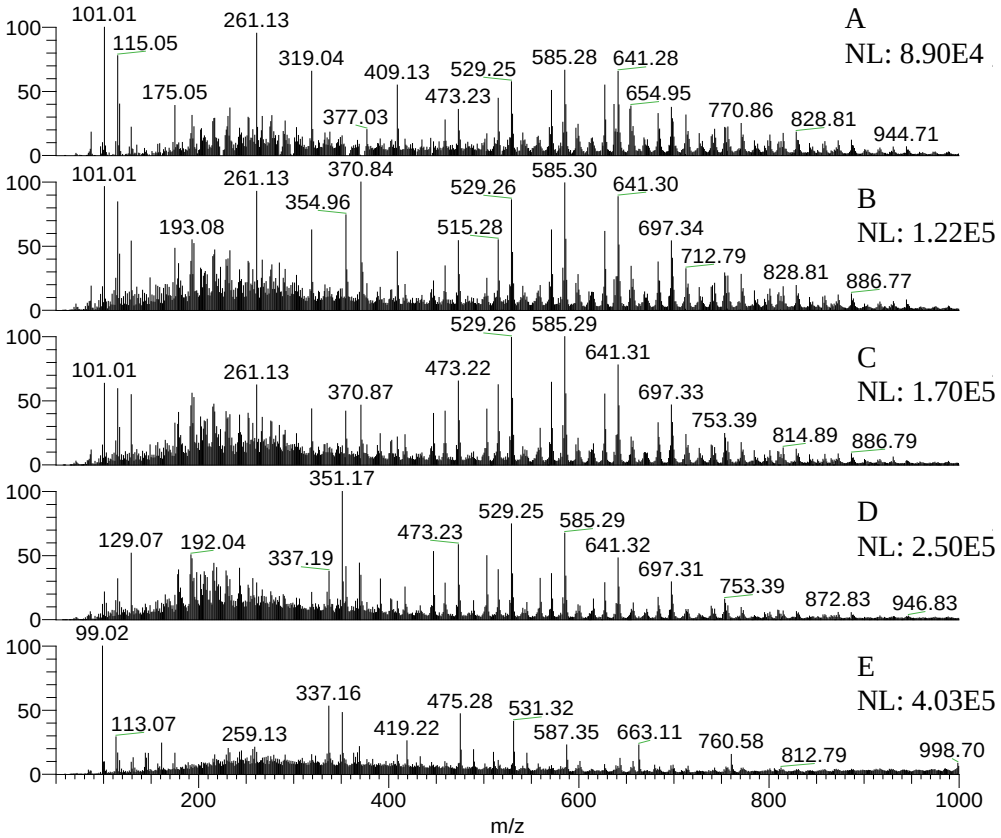

Figure 13. Neat Shell "set 1" gasoline analyzed at the optimized parameters; spectrum A corresponds to 0% evaporation, B to 25% evaporation, C to 50% evaporation, D to 75% evaporation and E to >90% evaporation.

As Figure 13 shows, there are characteristic high-mass ions in the form of a bell-curve and an overall CAS trend within a gasoline brand. Spectrum E of Figure 13 shows that some characteristic ions, such as m/z peak 697, can be lost in the >90% evaporation samples for the previously discussed reasons.

3.2.2 Neat Gasoline (Set 2) On QuickStrip^{TM} Cards Analyzed via DART-MS

Due to the effects of storage containers, involving the suspected leaching of compounds, and potentially the aging process of gasoline, a fresh set of gasoline samples was obtained from the same service stations. Fresh samples were acquired to ensure that unmodified gasoline could be analyzed effectively. The observed effects of storage

containers and aging are addressed later in the discussion. This set of gasoline was designated gasoline "set 2." These samples were analyzed under the same optimized parameters as gasoline "set 1;" however, the evaporation series and initial DART-MS neat sample analysis was performed on the same day the gasoline was purchased so that no time passed between the purchase and analysis of the fresh gasoline samples.

Only 0% and 50% evaporations were created with gasoline "set 2". These evaporation percentages were chosen because the 0% evaporation percentage in theory should contain all of the components that are initially present in the gasoline without losing them to evaporation. The 50% evaporation percentage was also chosen because it was previously observed to possess the most differentiation potential between brands, as shown by the analysis of gasoline "set 1". Additionally, the 50% evaporation maintains the higher-mass ions reflective of potentially discriminatory additives. The representative spectra of the 50% evaporated fresh gasoline obtained for the brands: Shell, Irving and Cumberland Farms are displayed in Figure 14.

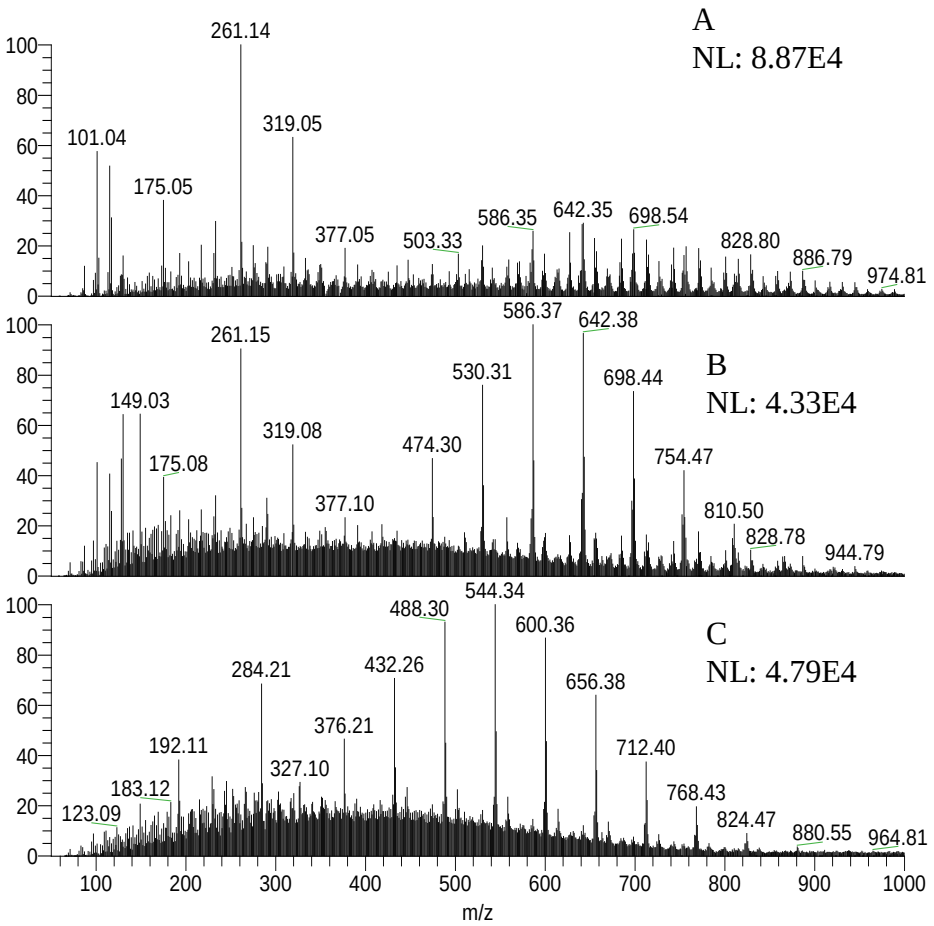

Figure 14. Gasoline "set 2," analyzed neat at 50% evaporation via DART-MS. Spectrum A corresponds to Shell, B to Irving and C to Cumberland Farms.

As shown in Figure 14, Shell and Irving share some similar ions. The Shell and Irving samples obtained in gasoline "set 2" were more similar to one another than these brands were observed to be in gasoline "set 1". Several of the same m/z peaks in the higher-mass region were detected in both brands, such as m/z peaks 586, 642 and 698. However, the apex of the characteristic "bell-curves" between the brands is still distinguishable, as are the less abundant ions existing between the peaks of the major bell-curve; Shell appears to have more additional lower-abundance ions in this mass range than Irving does. Cumberland Farms was distinguished from both Shell and Irving

through visual mass spectral analysis and was also observed to be similar to the Cumberland Farms gasoline sample from gasoline "set 1". Both Shell and Irving are similar to their respective spectra from gasoline "set 1", though less so than Cumberland Farms.

These samples were analyzed multiple times and were shown to be reproducible. The similarities between Shell and Irving could be attributed to a change in the refinery or previous distributor from which the gasoline was purchased since the first time the samples were acquired, or there could be a change in the additives based on the season. PCA confirmed that there were still differences, even if understated in the appearance of the mass spectral data, between Shell and Irving. Figure 15 shows a representative score plot created to determine if further assessment of the samples could be worthwhile.

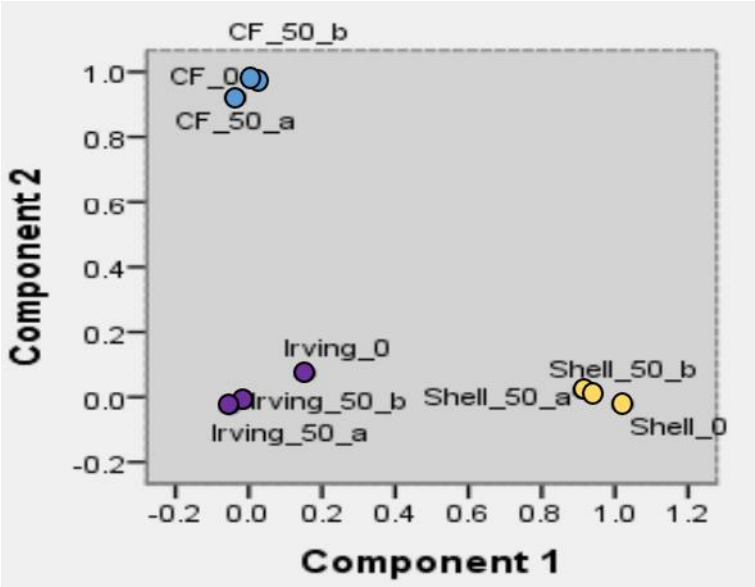

Figure 15. Score plot of the first two principle components in a 3-component PCA of both the 0% and 50% evaporation samples for Shell, Irving and Cumberland Farms (CF), duplicates of the 50% evaporation samples are present, designated by "_50_a" and "_50_b".

As the score plot shows in Figure 15, the Shell and Irving brands can be differentiated from one another by differences in the first two principal components. This result validates that the two brands can be differentiated despite sharing high-mass ion peaks in their mass spectral "bell-curves" and demonstrates that whole spectra must be taken into consideration for comparison and differentiation between gasoline brands. As shown in the score plot, all three brands were differentiated from one another. After verifying through PCA that the new "set 2" gasoline samples were distinguishable, dilutions were made from these samples. From these dilutions, the data for the chemometric models was subsequently collected.

After the successful differentiation of the "set 2" gasoline samples among the first three brands, new samples of the remaining two brands, Sunoco and Gulf, were also purchased for fresh sample analysis. These samples were purchased from the same service stations as those in gasoline "set 1". These two brands were also purchased slightly longer than 1 month after the purchase of the gasoline "set 2" samples of the Shell, Irving and Cumberland Farms brands, leaving open the possibility of further seasonal variability. 0% and 50% evaporation samples were prepared on the same day as purchase and subsequently analyzed in replicates.

Both 0% and 50% evaporated Sunoco and Gulf samples looked highly similar to one another, as shown by Figure 16. These "set 2" Sunoco and Gulf neat samples were dissimilar from the "set 1" Sunoco and Gulf neat samples that had been analyzed previously. The difference observed between gasoline "set 1" and "set 2" was hypothesized to be due to service stations purchasing from a different refinery or

distributor, or the service stations implementing the use of seasonal summer additive blends that were different than those used in gasoline "set 1". Due to proprietary limitations, this hypothesis was not able to be confirmed.

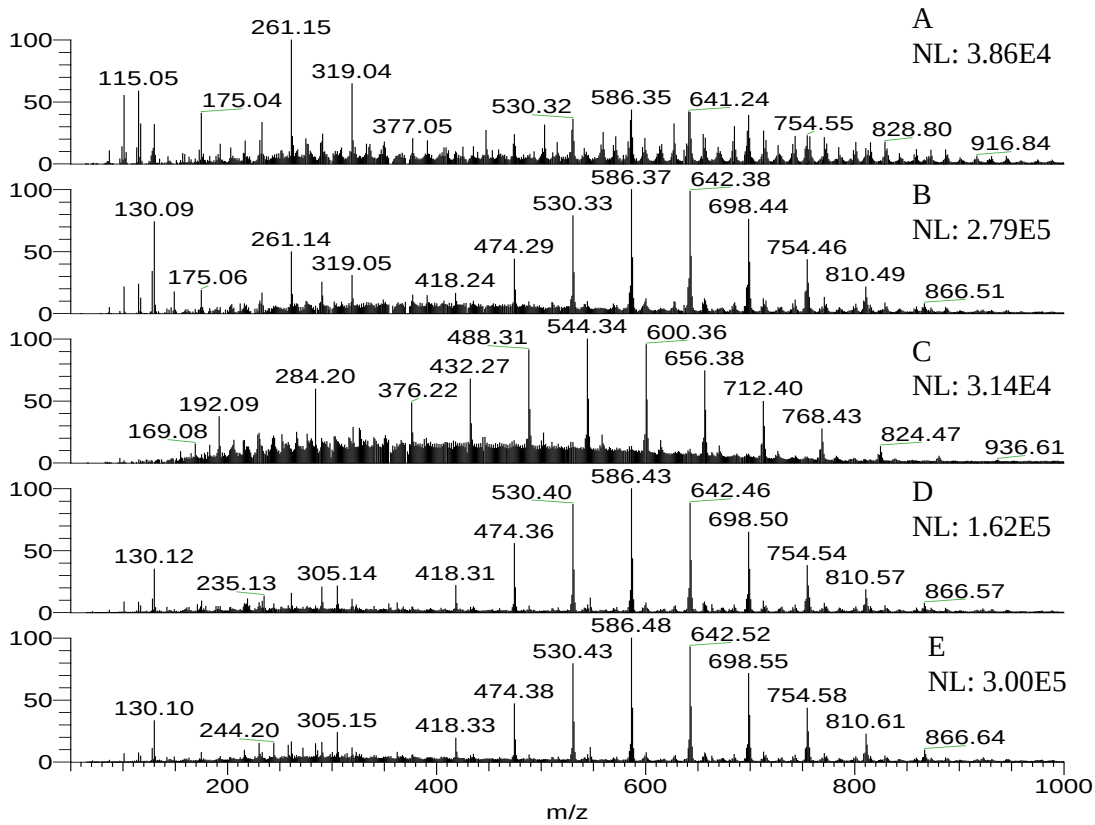

Figure 16. Demonstration of the significant mass spectral similarities between Sunoco, Gulf and Irving. A corresponds to Shell, B to Irving, C to Cumberland Farms, D to Sunoco and E to Gulf.

The resulting Sunoco and Gulf spectra were observed to not only be similar to one another, but to the gasoline "set 2" Irving sample, demonstrated by Figure 16. These observations lead to concerns regarding whether or not mass spectral and chemometric differentiation would be able to be achieved.

3.2.3 Neat Gasoline (Set 1) On Carbopack™ X Analyzed via DART-MS

Carbopack™ X is a sorbent-coated mesh that is coated with graphitized carbon. Different sorbents have varying affinities for different analytes of interest. This sorbent was utilized by another researcher in the same research group for the analysis of smokeless powders, which was successful. Preliminary experiments were performed to determine if this sorbent-coated mesh would also have an affinity for the gasoline analytes. The same procedure was employed as the one used with the neat gasoline analysis on QuickStrip™ cards, which is described in the experimental design section *2.4.1 Neat Gasoline Sample Analysis*. The objectives were to utilize Carbopack™ X as the target analyte collection material and determine if it performed better than the QuickStrip™ cards. The intent was that if it produced favorable results with the neat gasoline samples, including high ion count signals and detection of the CASs for each gasoline, then it could serve as a better collection material than QuickStrip™ cards with diluted gasoline samples. As stated previously in this research, initial attempts at diluting the gasoline samples generated low signal ion counts when analyzed with the QuickStrip™ cards. The results of the gasoline analysis with Carbopack™ X are shown in Figure 17.

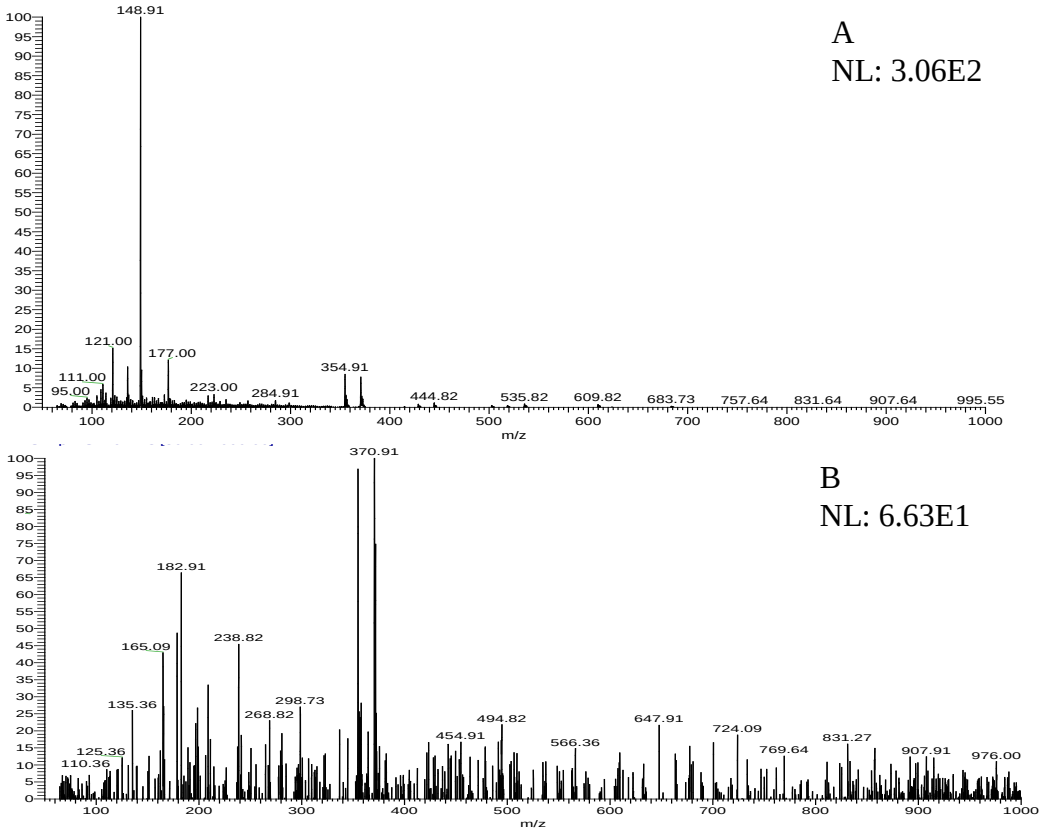

Figure 17. Neat 50% evaporated Shell spotted onto Carbopack™ X. Spectrum A corresponds to the Carbopack™ X blank, and spectrum B corresponds to the Shell sample; both were held with forceps in front of the ionization source.

As the background Carbopack™ X spectrum, shown in Figure 17A, and the resulting gasoline spectrum, Figure 17B, show, the target analytes of interest and related characteristic patterns were not able to be detected with this sorbent. All of the brands and evaporation percentages at various ionization temperatures analyzed resulted in similar spectra. Even with background subtraction, the sample spectra were overwhelmed by background ions, and the sample ion signals were exceptionally low, as shown by the 6.63 x 10 ion counts observed in Figure 17B. This sorbent-coated mesh was not used in future experiments due to its low affinity for gasoline analytes or potential lack of desorption of target analytes from the mesh.

3.2.4 Analysis of Storage and Age Effects on Gasoline

After acquiring and assessing all of the resulting spectra for the neat gasoline samples for each brand, various solvents were evaluated in an effort to analyze diluted samples, which are commonly encountered in forensic case-work. After creating a new Shell evaporation series from gasoline "set 1", dilutions were made in pentane, carbon disulfide and methylene chloride. These three different solvents were investigated to determine which solvent was most amenable to generating the characteristic spectra observed for the neat gasoline samples. However, during the solvent evaluation, a trend was noticed with gasoline samples "set 1", including all of the evaporation series created from this set. Prominent m/z peaks 510 and 663 were observed to be present in all of the Shell evaporation series from "set 1". The 663 peak was noticeably more abundant than the characteristic m/z peaks of the signature high-mass ions in the form of a bell-curve, as well as many of the other m/z peaks present in the original neat samples. The previously observed intense CAS peaks were reduced to the baseline while the m/z 510 and 663 peaks dominated the spectra. The new and old evaporation series, both neat and diluted from gasoline "set 1", were analyzed for all of the five gasoline brands to confirm that the trend was present in all of the brands. Solvent blanks of pentane, carbon disulfide and methylene chloride were also analyzed in efforts to determine the source of the m/z 510 and 663 peaks.

The increasing abundance of the m/z 510 and 663 peaks is apparent from the three evaporation series created from gasoline set 1 on 11/10/14, 12/05/14 and 3/11/15, shown in Figure 18. This effect was observed in both the neat and the diluted samples.

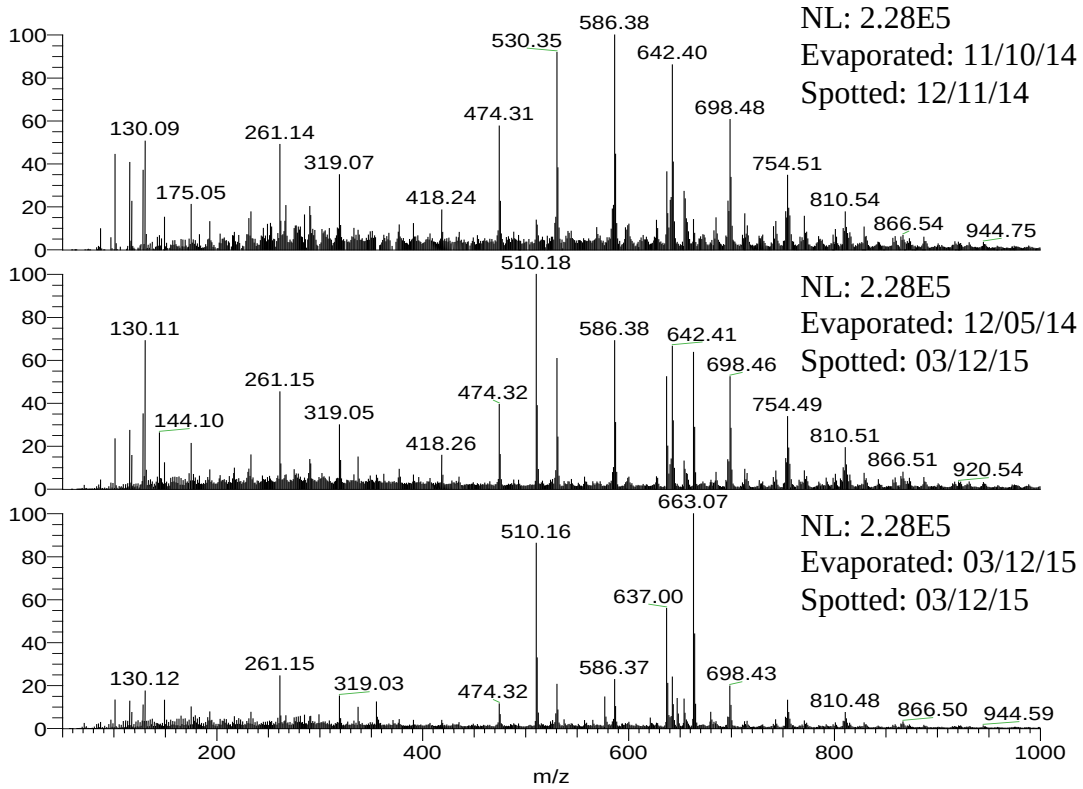

Figure 18. Neat 50% evaporated Irving samples from various evaporation dates and analysis dates, all from gasoline "set 1".

This trend posed concern for the continued use of this specific set of gasoline samples because the characteristic ions were becoming more difficult to detect. To determine if this effect was present in only the Shell brand of gasoline, all of the other brands were subsequently analyzed in the same manner. This effect was demonstrated to be present in all five of the brands and was exacerbated with diluted samples. As demonstrated by Figure 18, the newer evaporation samples were increasingly overwhelmed by m/z 510 and 663. This trend was not only noticed with newly created evaporation samples, but with the spotting of the original evaporated samples Due to concern about how the resulting spectra may appear with increasing age, it was

determined that new gasoline samples should be obtained before additional analyses were conducted.

To further investigate the source of the m/z 510 and 663 peaks, a new plastic gasoline container was purchased, a section of the plastic was excised and then analyzed under the same optimized DART-MS parameters as the gasoline samples.

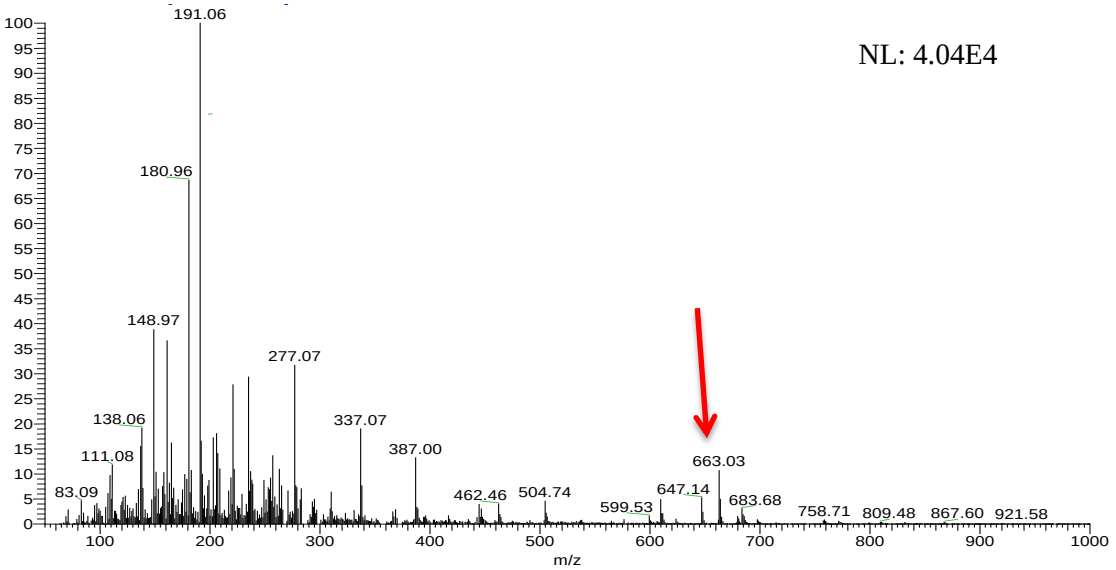

Figure 19. DART-MS spectrum of gasoline container plastic, with background subtraction.

The m/z 663 peak, designated by the red arrow, can be observed in the spectrum of Figure 19. While the m/z 663 peak is present, it is not the most abundant peak. Other ions are observed that were not detected in the gasoline samples and are likely other components of the plastic. From this spectrum, it was hypothesized that as the gasoline remained in the container for months, this compound was preferentially leached into the gasoline, likely due to this compound having similar properties to the gasoline, and it accumulated over time. The m/z 663 peak was also observed in far lower abundances in

the original spectra of several of the gasoline brands. Even then, it could have been preferentially leached from the container, but to a much lesser extent due to the gasoline not having resided in the container for as long. However, the m/z 510 peak was not observed in the plastic container spectrum, as seen in Figure 19. It was therefore hypothesized that more than one factor was contributing to this effect in the gasoline samples. After subsequent literature searches, it was postulated that the gasoline was aging, and the relative abundances of specific groups of compounds were decreasing at faster rates than others. Many environmental associations have developed methods for determining the age of gasoline for reasons such as investigating the age of a gasoline spill or leak into ground water. It is likely that the leaching of the m/z 663 peak occurred simultaneously with the aging effects as all of the gasoline brands remained in their respective gasoline containers. These results confirmed that it was necessary to purchase fresh gasoline samples, which comprised gasoline "set 2". Other literature of ignitable liquid studies noticed the aging of ignitable liquids as well, noting that the density of fuel samples is observed to increase as fuel ages due to the evaporation of lighter fractions and gum formation[21].

3.2.5 Solvent Analysis

After acquiring fresh gasoline samples, solvent evaluation was performed again with pentane, carbon disulfide and methylene chloride on the new Shell, Irving and Cumberland Farms gasoline samples. While the CASs respective of each of the three brands was detectable with dilution in each solvent, there were a few minor differences between the resulting spectra of each sample diluted in the three solvents. The ion counts

for the samples diluted with carbon disulfide and methylene chloride were slightly higher than the ion signals of the resulting sample spectra diluted in pentane. For example, 50% evaporated Shell from gasoline "set 2", had ion counts of 3.04×10^4 and 3.94×10^4, respectively for carbon disulfide and methylene chloride, and a lower signal of 1.12×10^4 for pentane. The CASs were better observed in dilutions of carbon disulfide and methylene chloride than in pentane, though not substantially. However, no dramatic differences were observed between the samples diluted in the three solvents, and the characteristic ion peaks were detected during analysis with all three solvents. Despite pentane producing slightly lower ion counts and less obvious characteristic mass spectral patterns, pentane was chosen as the solvent to use in the additional experimentation because pentane is a much safer and easier solvent to work with. Additionally, pentane is a common solvent used in fire debris analyses currently performed in laboratories, so it would be convenient to use pentane in additional gasoline analyses.

After choosing pentane as the solvent, fresh dilutions were made for all five brands of the gasoline "set 2" samples. The CASs were still detectable, and the mass spectra of the diluted samples could still be visually differentiated. The high-mass ions composing the characteristic bell-curve of each brand were also still observed despite dilution, as demonstrated in Figure 20.

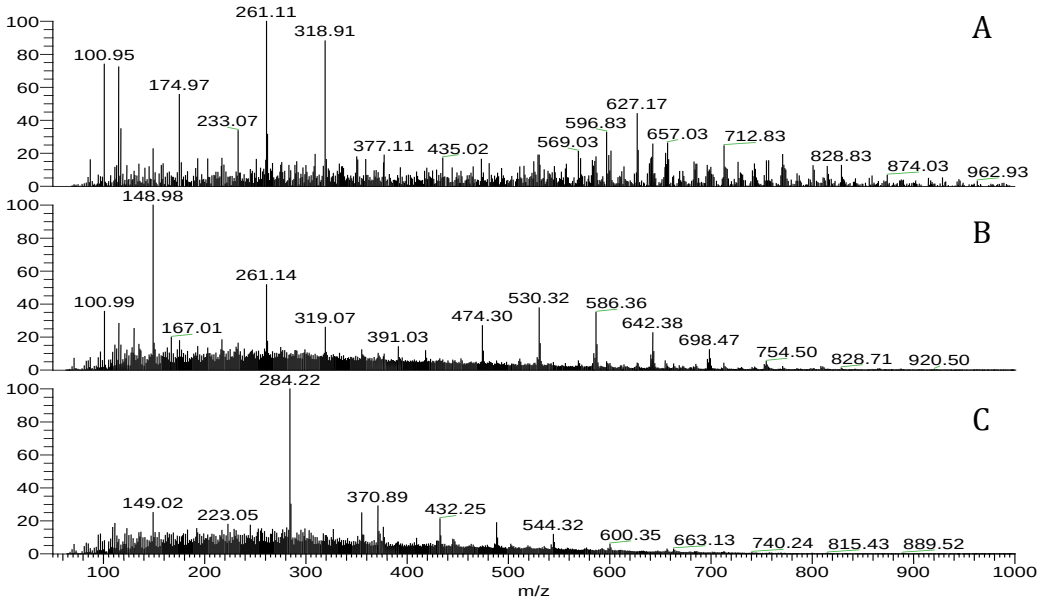

Figure 20. 50% evaporation samples in a 1:10 pentane dilution, gasoline "set 2", spectrum A corresponds to Shell, B to Irving and C to Cumberland Farms.

The majority of high-mass ions composing the characteristic bell-curve of each brand were also still observed; however, the abundances of some of the higher-mass ions of the characteristic bell-curves appeared to shift, causing a different ion to be at the apex of the curve in comparison to the undiluted samples. This effect likely occurred because the higher-mass ions with lower abundances were not detected due to the higher dilution factor. The ion counts from the 1:25 dilutions were not as intense as those observed with the 1:10 dilutions; however, there was still a noticeable signal and characteristic patterns in the spectra at both dilutions.

3.3 Comparative GC-MS Spectra of Gasoline

Traditional methods of analyzing fire debris, especially gasoline, involve sample evaluation utilizing GC-MS. While this method is sensitive and able to differentiate

between certain degrees of gasoline weathering, traditional methods do not offer brand differentiation or generation of higher-mass ions for gasoline samples.

To compare the gasoline spectra generated from DART-MS analysis with spectra obtained from traditional methods, the gasoline samples were analyzed using heated passive headspace concentration in accordance with ASTM guidelines followed by subsequent GC-MS analysis[11,23]. Gasoline "set 1" samples were extracted with the heated passive headspace method and then desorbed with pentane before being analyzed. Figure 21 displays all of the brands at the 50% evaporation, from gasoline "set 1," analyzed by GC-MS.

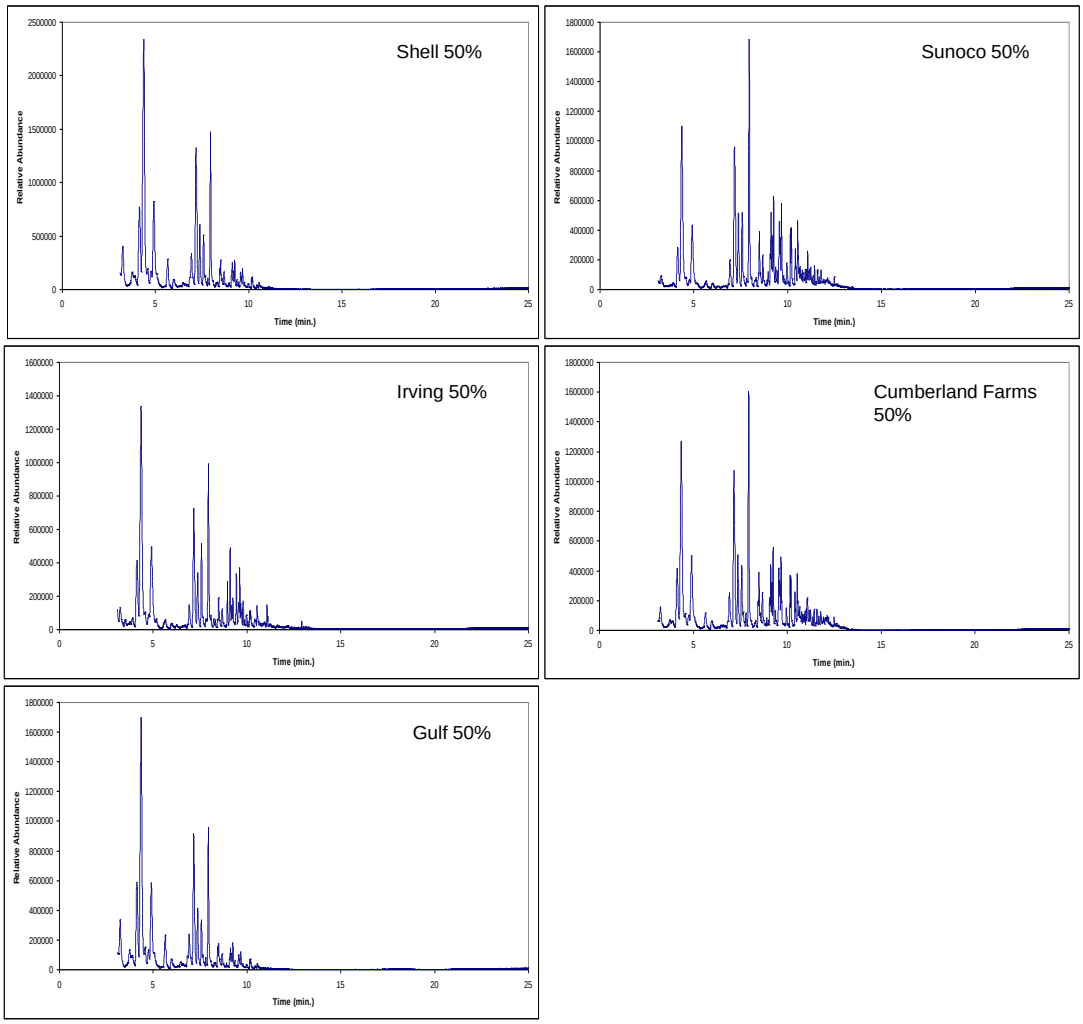

Figure 21. Chromatograms generated from all five brands of gasoline at 50% evaporation, gasoline "set 1", utilizing a traditional heated passive headspace concentration method followed by GC-MS analysis.

As demonstrated in Figure 21, other than a few differences in relative abundances, no significant chromatographic differentiation between gasoline brands is observed when samples are analyzed utilizing GC-MS preceded by heated passive headspace extraction. Each of the above spectra is characteristic of gasoline, but no CASs were detected with this analytical approach.

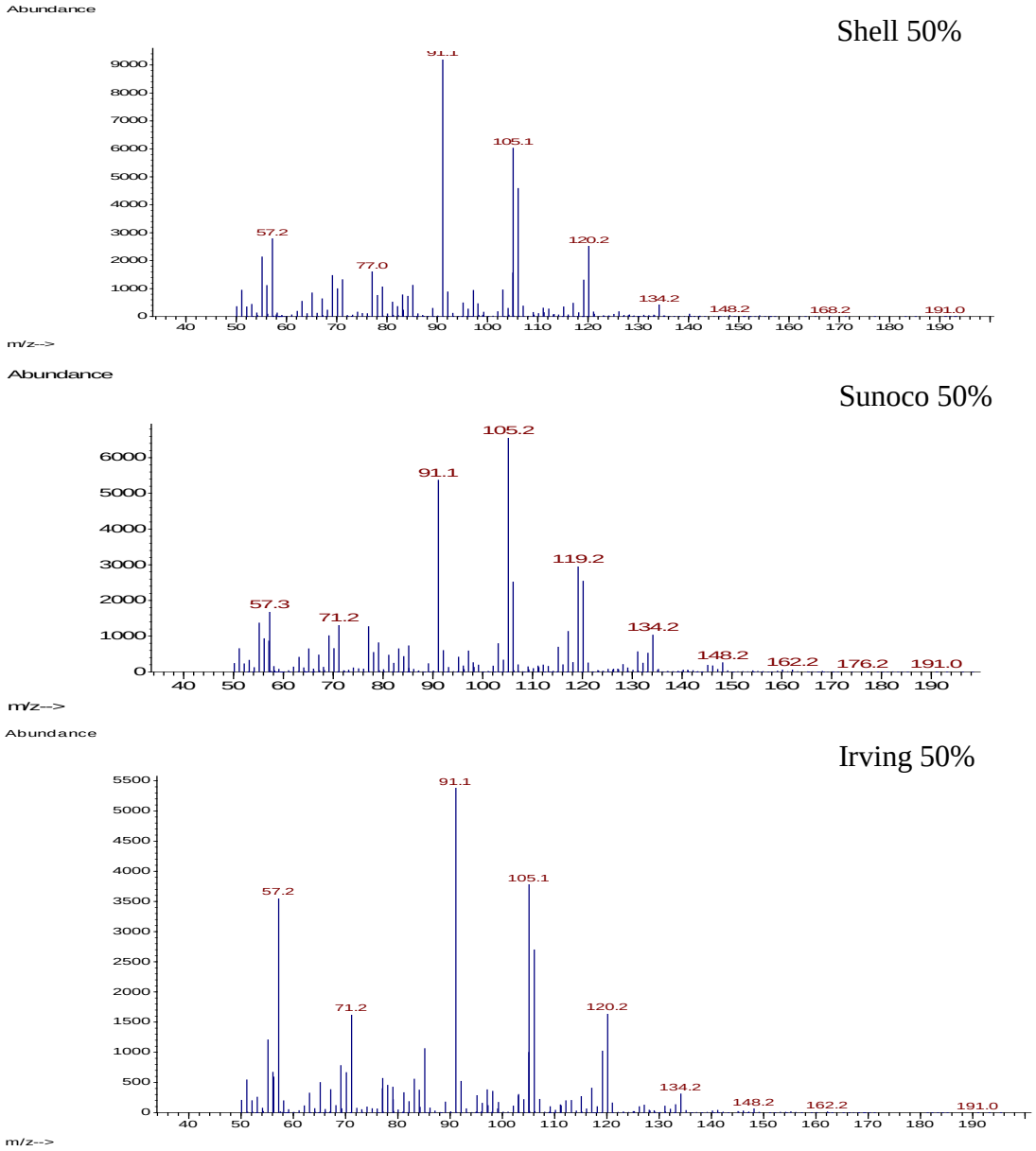

Figure 22. Summed ion data for Shell, Sunoco and Irving brand gasoline, "set 1," at 50% evaporated, generated utilizing traditional heated passive headspace and GC-MS methods.

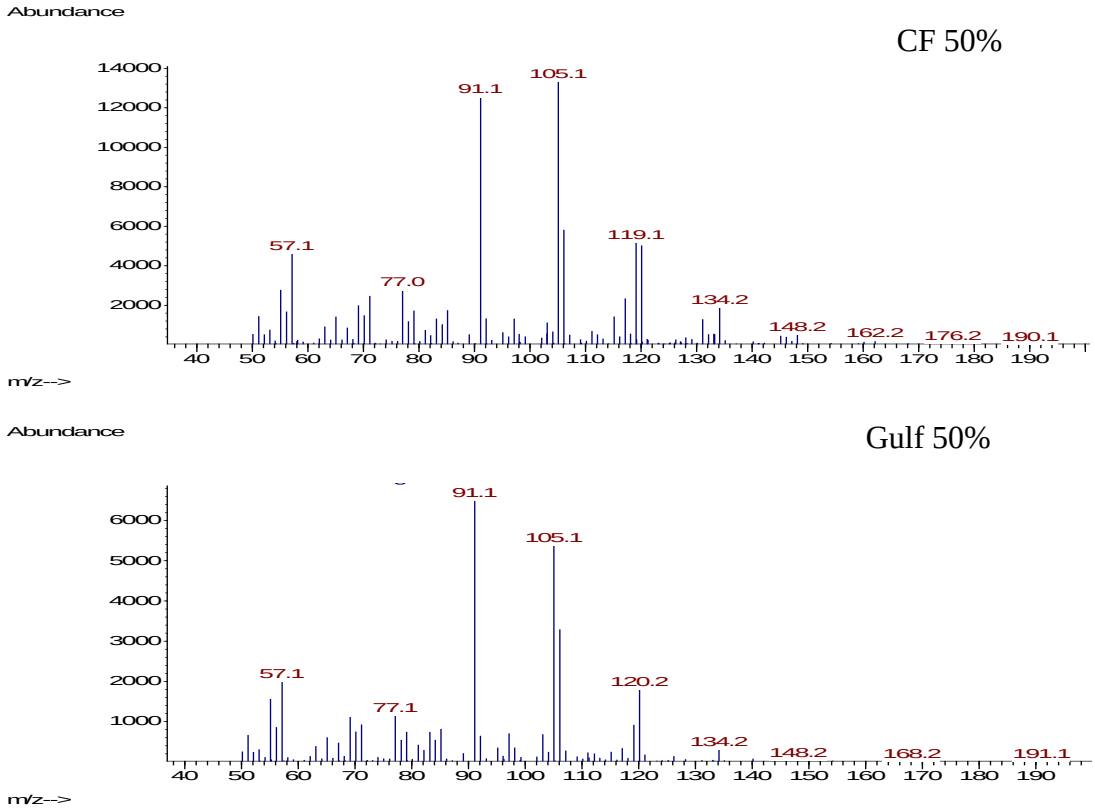

Figure 23. Summed ion data for Cumberland Farms and Gulf brand gasoline, "set 1," at 50% evaporated, generated utilizing traditional heated passive headspace and GC-MS methods.

As shown in Figures 22 and 23 above, there are no significant differences in the summed ion mass spectral data between the five brands of gasoline in "set 1" at the 50% evaporation. The summed ion data was generated by summing the total mass spectra across the total ion chromatogram (TIC) obtained for each gasoline sample; the spectra show all the mass spectral data spanned over the total collected retention times in the gas chromatograms of each sample. Additionally, no mass spectra rich in higher-mass ions, or any ions with m/z greater than approximately 134 were detected in significant abundance, as shown in Figures 22 and 23; however, it should be noted that the samples were not scanned above 550 amu, shown by Table 4, and that the images were cropped

for clarity due to no significant m/z peaks being present past the values displayed. Overall, no differentiation between gasoline brands or detection of brand-specific CASs was possible with the traditional GC-MS methods used by many crime labs today.

The same GC-MS procedures and data collection were performed with gasoline "set 2" samples for the 0% and 50% evaporations. Again, no significant differentiating characteristics were observed among the five brands of gasoline for either the gas chromatograms or the summed ion data derived from them.

3.4 Identification of High-Mass Ions

To better understand the observed CASs of the five brands generated by DART-MS, efforts were made to identify the higher-mass ions and subsequently associate them with possible gasoline additives. As noted previously, many additives are intentionally added by manufacturers of the various brands of gasoline to allow for identification of a specific gasoline source in instances such as environmental contamination and gasoline spills[22]. These identifying additives are also likely added by each company for commercial purposes as well, serving as positive control additives[48]. The peaks of the signature "bell-curves" associated with each brand of gasoline, in both "set 1" and "set 2," were found to have a trend in the mass differences between each major peak present.

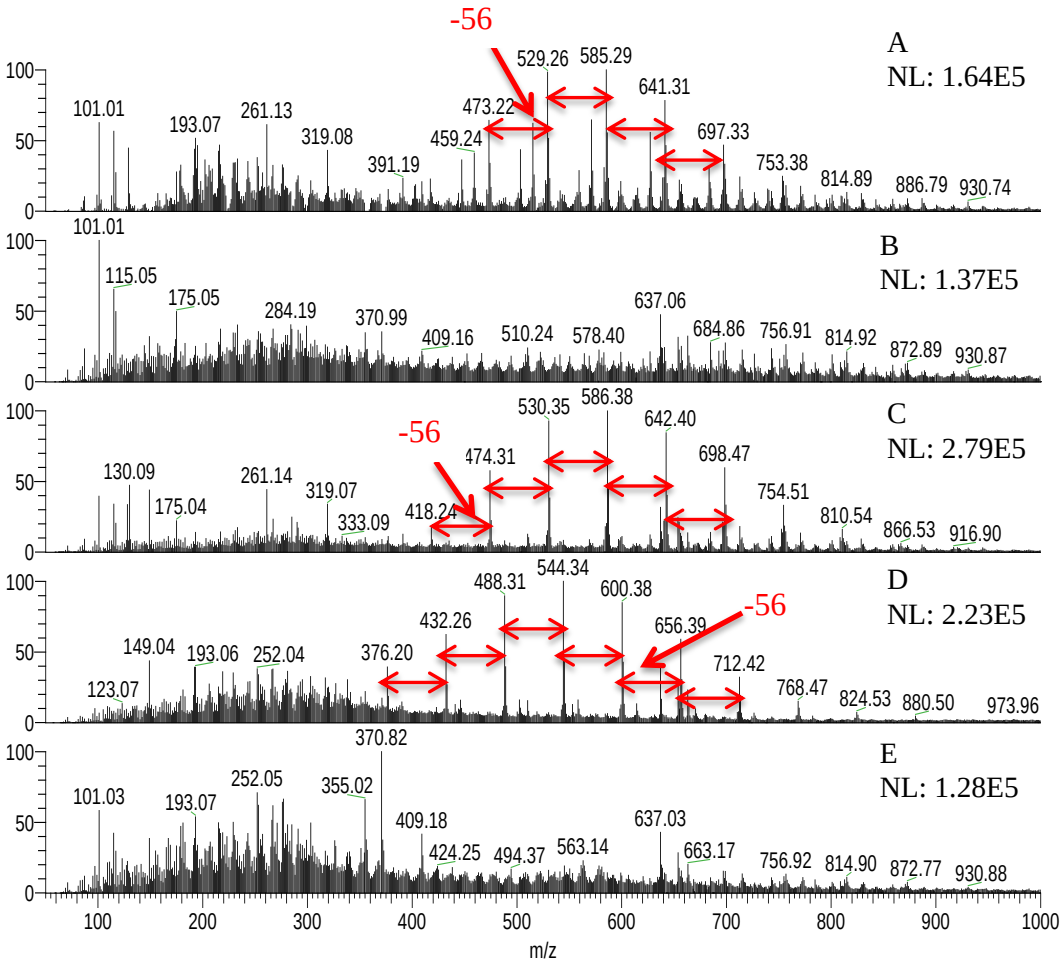

Figure 24. All of the "set 1" gasoline brands analyzed by DART-MS, neat at a 50% evaporation; spectra A corresponds to Shell, B to Sunoco, C to Irving, D to Cumberland Farms and E to Gulf. The repeating unit of 56 amu between the higher-mass ions is designated with arrows.

As shown in Figure 24, three of the five brands of gasoline in "set 1", Shell, Sunoco, and Cumberland Farms, were found to have a repeating difference between the higher-mass ions in their characteristic "bell-curves" of 56 amu. While there were higher-mass ions observed in Sunoco and Gulf brand gasoline, no repeating trend was found between them.

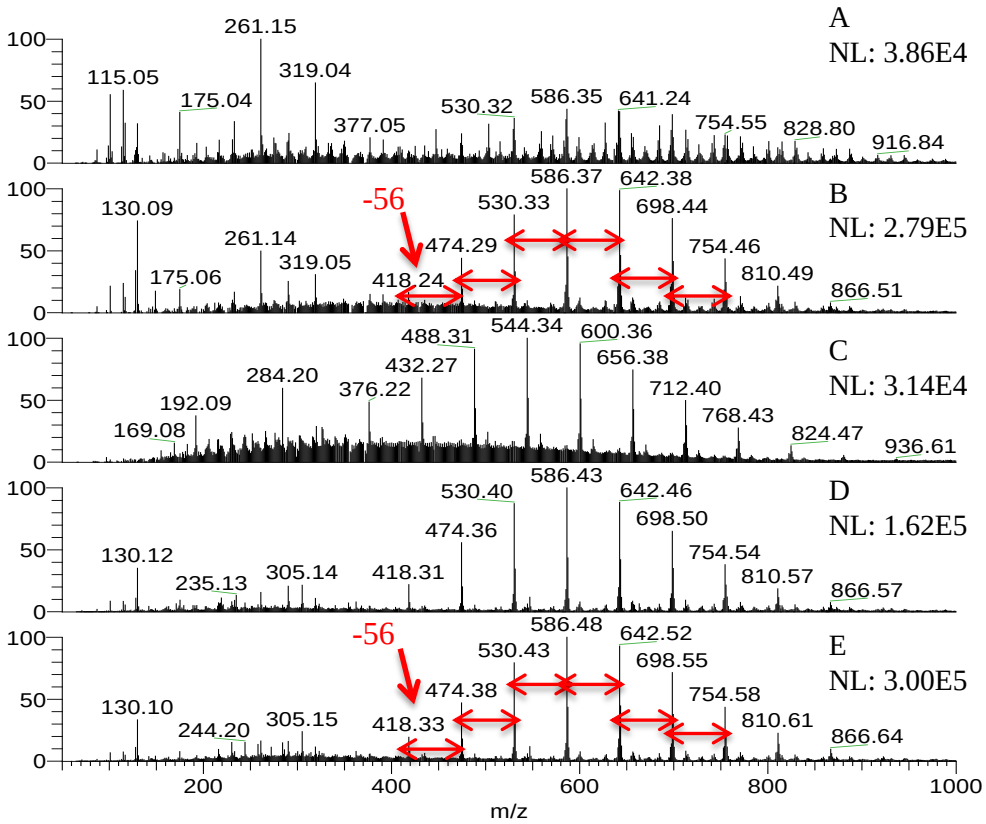

Figure 25. All of the "set 2" gasoline brands analyzed by DART-MS, neat at a 50% evaporation; spectra A corresponds to Shell, B to Irving, C to Cumberland Farms, D to Sunoco and E to Gulf. The repeating unit of 56 amu between the higher-mass ions is designated with arrows.

As in gasoline "set 1," in gasoline "set 2," the Shell, Irving and Cumberland Farms gasoline brands, at a 50% evaporation, were again found to have a consistent repeating trend between the higher-mass ions in their CASs. Additionally, the Sunoco and Gulf brands, evaporated to 50%, were also found to have a repeating 56 amu difference between the higher-mass peaks in gasoline "set 2". This observed trend in mass differences between higher-mass ions was also observed in a study performed on different brands of gasoline from a different region[48]. This study attributed these observations to polymeric gasoline additives and noted that polymeric additive packages

were observed to differ between gasoline brand and region[48]; this referenced study did not analyze the same brands as those investigated in this research. These higher-mass ions, therefore, likely represent polymeric repeats[48]. While the repeating unit was determined to be the same for all five gasoline brands, it is important to note that CASs are still present based on the overall mass spectral patterns and observable differentiations. Also, while the polymeric repeating unit is the same between the brands, the actual polymer itself, including any associated functional groups, could still be different between the brands. These observations support that the higher-mass ions detected during the course of this research are true peaks present in gasoline.

3.5 Chemometric Analysis

After analyzing both gasoline "set 1" and "set 2", neat, utilizing DART-MS and traditional heated passive headspace extraction followed by subsequent GC-MS, all of the data was processed by the methods listed in the experimental design section, and score plots were created for each analysis method to show comparisons between the two methods.

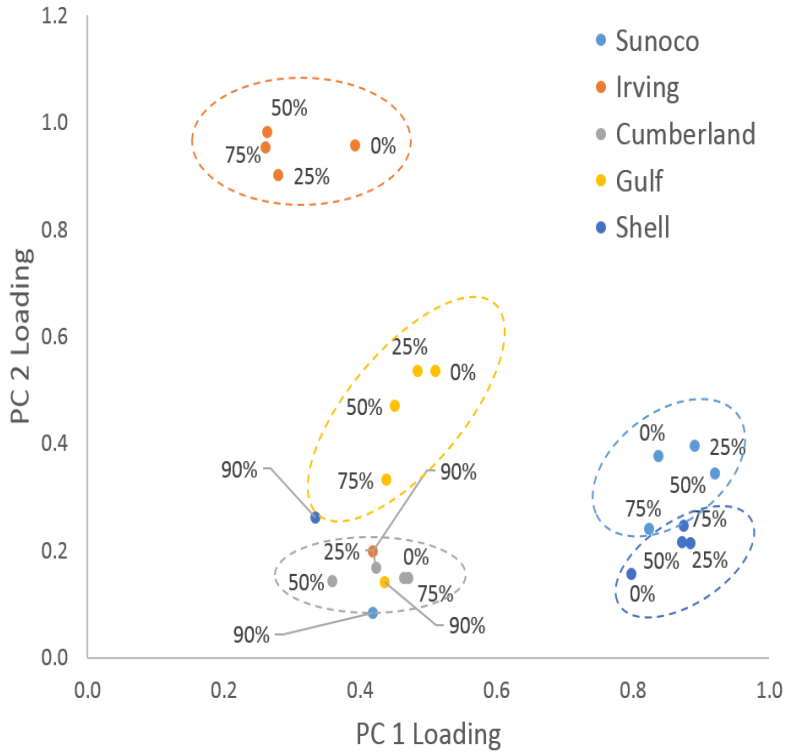

Figure 26. Each brand of gasoline "set 1," analyzed at all evaporation percentages utilizing DART-MS. This score plot is courtesy of Dr. Matthew Pavlovich.

Data in the score plot shown in Figure 26 was generated from mass spectral data acquired by DART-MS analysis of gasoline "set 1" samples. As demonstrated in Figure 26, each gasoline brand is observed to cluster separately from one another. The >90% evaporated samples are an exception to this observation and appear to cluster closer to one another than to their respective brands.

Conversely, the same gasoline samples at the same evaporation percentages are not observed to cluster based on brand when analyzed using traditional GC-MS methods.

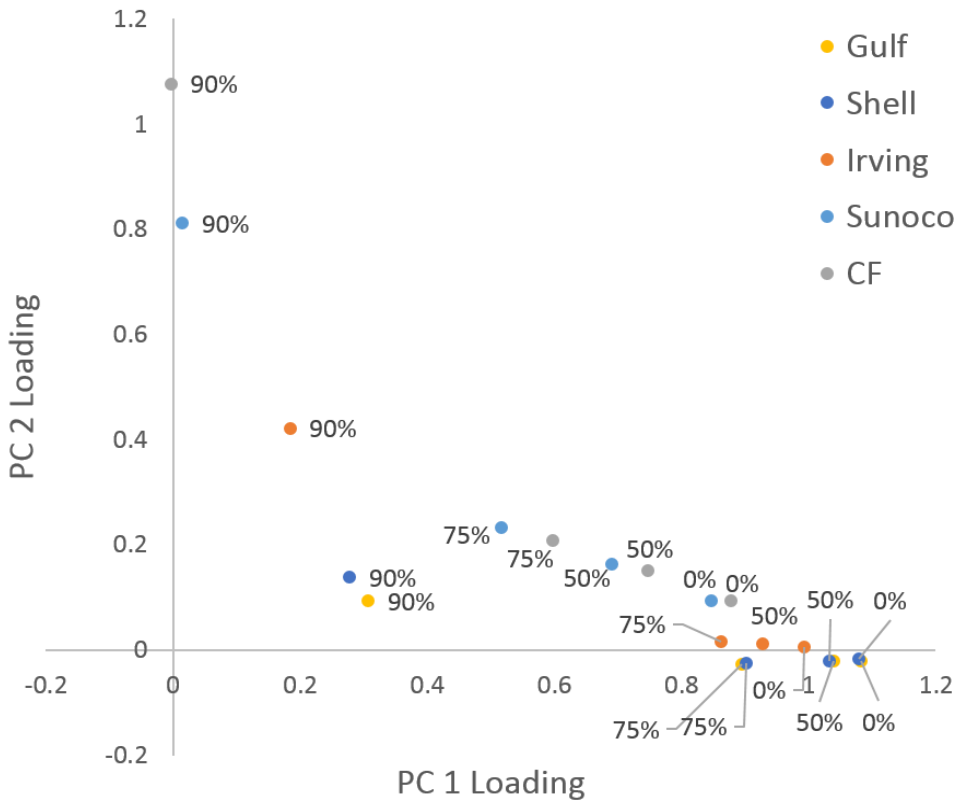

Figure 27. Clustering of gasoline "set 1" samples at all brands and evaporation percentages generated utilizing GC-MS methods. This score plot is courtesy of Dr. Matthew Pavlovich.

The score plot in Figure 27 was generated from chromatographic data derived through heated passive headspace extraction followed by GC-MS analysis of gasoline "set 1" samples. As demonstrated in Figure 27, the samples are not observed to cluster based on brand, indicating that the differentiation power achieved with DART-MS analysis is superior to that of GC-MS analysis for classifying gasoline samples based on brand.

A score plot was also generated for gasoline "set 2" data. The neat data was utilized for this analysis to see the inherent differentiation between the five brands prior to any procedural alteration of the samples. Both 0% and 50% evaporations were analyzed for generation of this score plot; duplicate spectra were included for each brand at each of the two evaporation percentages. No background subtraction was performed on the samples inputted into the software that was used to generate this score plot

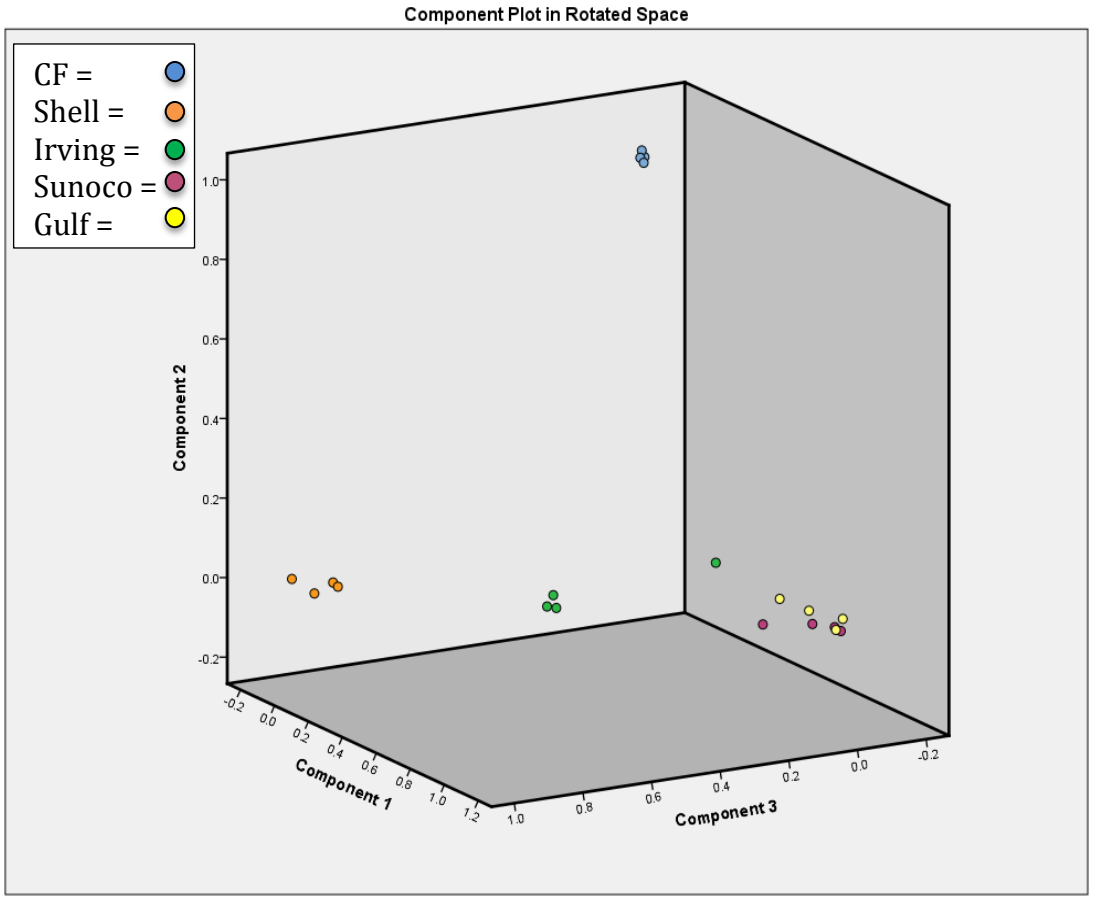

Figure 28. Clustering of neat gasoline "set 2" samples of all brands at 0% and 50% evaporations (analyzed in duplicate) generated utilizing DART-MS methods; associated variances also included. This plot is courtesy of Dr. Matthew Pavlovich.

As shown by the score plot displayed in Figure 28, all five brands are mostly differentiated from one another utilizing DART-MS analysis. It is important to note that this differentiation was still achieved even with the differences observed between gasoline "set 1" and "set 2," as well as the striking similarities observed between Sunoco, Irving and Gulf brands in "set 2", which can be seen spectrally in Figure 16.

Table 8 shows the models constructed for each of the five gasoline brands using chemometric analysis. No background subtraction was performed on any samples collected for the chemometric model construction due to the sample acquisition method within the software. However, the appropriate blank samples referenced in section *2.6.2 Analyze IQ and Model Building* were utilized to train the models with samples that were not gasoline to prevent possible misclassifications. Spectral attribute voting with the "normalize" preprocessing method was observed to generate the models with the lowest error rates for each of the gasoline brands. All of the models generated training error rates below 5%. As Table 8 shows, testing error rates as low as 0% were obtained for some of the gasoline brands, such as Shell. All of the models produced zero false positives except one Sunoco model and one Gulf model. However, these two models were only trained with two gasoline brands. All of the models trained with all five brands produced zero false positives. This result demonstrates that the classification power increases as more sample types are added to the models. The number of false positives generated by a forensic approach is an important value to consider in method development because laboratory work is often linked to a potential legal consequence. To construct models able to accurately classify unknown gasoline brands, models should

be trained with a variety of samples from various brands, service stations, geographical locations, seasons and degrees of aging.

Table 8. Gasoline model parameters and results for all five brands using spectral attribute voting.

Gasoline Brand	*Model*	*# of Total Samples Trained With*	*# of Brands Trained With*	*Pre-processing: Normalize*	*Training Error*	*Testing Error*	*# of False Positives*	*# of False Negatives*
Shell	SAV	908	3	Yes	0.22%	0%	0	0
Shell	SAV	1488	5	Yes	0.52%	0%	0	0
Irving	SAV	908	3	Yes	0.48%	1.34%	0	2
Irving	SAV	1488	5	Yes	1.03%	5.37%	0	7
CF	SAV	908	3	Yes	0.11%	0.67%	0	1
CF	SAV	1488	5	Yes	0.23%	5.37%	0	8
Sunoco	SAV	579	2	Yes	0.76%	4.70%	7	0
Sunoco	SAV	1488	5	Yes	0.83%	0%	0	0
Gulf	SAV	579	2	Yes	1.14%	4.03%	5	0
Gulf	SAV	1488	5	Yes	0.70%	5.37%	0	0

3.6 Blind Study

Fifty diluted samples were created from the five gasoline brands using the 0% and 50% evaporations. The samples were prepared blindly to the analyst and subsequently analyzed via DART-MS under the optimized parameters. The blind samples were analyzed and resulting data was inputted into the Analyze IQ Lab software in the same manner as both the training and testing data sets. Again, no background subtraction was performed on these samples due to the data acquisition method. The blind samples were tested against the five gasoline brand models that were trained with all five gasoline brands, pentane blanks, QuickStrip™ cards and ambient background blanks. After the prediction classifications were generated for the blind data set by the Analyze IQ Lab

software, the true identity of the unknowns were recorded to deduce the error percentages of the Shell, Sunoco, Irving, Cumberland Farms and Gulf models.

As the samples were analyzed blindly utilizing DART-MS, it was observed that the m/z 663 peak was present in a majority of samples. As noted previously in section *3.2.4. Analysis of Storage and Age Effects on Gasoline*, this peak was determined to greatly reduce the signal of the peaks representative of the CASs of each brand due to changes in relative abundances in comparison to non-aged, or fresh, gasoline samples. As shown in Table 9, the error percentages of the five trained gasoline models tested against the blind data set were significantly high, with only two brands, Shell and Cumberland Farms having a percent error rates below 50%.

Table 9. Details and results of the blind study including percent error derived from the DART-MS analysis of 50 unknown diluted gasoline samples; CF represents Cumberland Farms.

Brand	*Evaporation %*	*Dilution*	*Total # of Samples*	*# of False Negatives*	*Total # Correctly Classified*	*Total # of Misclassi -fications*	*% Error*	*Total Brand % Error*
Shell	0%	1:10	6	1	5	1	17%	40.74 %
		1:25	6	5	1	5	83.33 %	
	50%	1:10	9	0	9	0	0%	
		1:25	6	5	1	5	83%	
Sunoco	0%	1:10	6	6	0	6	100%	88.89 %
		1:25	6	6	0	6	100%	
	50%	1:10	9	6	3	6	33%	
		1:25	6	6	0	6	100%	
Irving	0%	1:10	6	6	0	6	100%	73.08 %
		1:25	6	6	0	6	100%	
	50%	1:10	8	1	7	1	87.50 %	
		1:25	6	6	0	6	100%	
CF	0%	1:10	6	3	3	3	50%	40.74 %
		1:25	6	6	0	6	100%	
	50%	1:10	9	1	8	1	88.89 %	
		1:25	6	1	5	1	17%	
Gulf	0%	1:10	6	6	0	6	100%	100%
		1:25	6	6	0	6	100%	
	50%	1:10	9	9	0	9	100%	
		1:25	6	6	0	6	100%	
Pentane	NA	NA	15	0	15	0	0%	NA
Totals	*NA*	*NA*	*149*	*92*	*57*	*92*	*61.70 %*	*68.66 %*

While the percent errors for this data set were significantly high and unfavorable, it is plausible that the high number of misclassified samples are attributable to the presence of the m/z 663 peak in a majority of the mass spectra in addition to any other spectral changes caused by aging that caused the CASs to be less similar to the training data. It is important to note that the gasoline "set 2" samples had been stored in glass, as opposed to the plastic containers that the gasoline "set 1" samples were stored in, yet the m/z 663 peak was still present, indicating that this is likely a change resulting more from effects of aging than from the storage container. Building a model incorporating both fresh and aged gasoline for each brand could improve the classification predictions and percent errors of the models. It is possible that if the blind study was conducted at an earlier date, the false negatives would not have been as prevalent.

Lastly, it is important to notice in Table 9 that none of the blind pentane samples were classified as any of the gasoline brands. Also, none of the gasoline brands were misclassified as any of the other brands. This result demonstrates that no false positives were produced in the entire blind study, only false negatives, which is an important factor from a forensic analysis standpoint.

4. CONCLUSIONS

In conclusion, this research has demonstrated that gasoline brands can be differentiated utilizing a DART-MS analytical approach. Chemical attribute signatures were generated based on brand for both neat gasoline samples and samples diluted in various solvents including pentane, carbon disulfide and methylene chloride via DART-MS. Differentiation can also be observed within one brand with various evaporation

percentages, or degrees of weathering; however, an overall trend in the general mass spectral pattern was demonstrated between samples of the same brand. For gasoline, this DART-MS analytical approach exhibits higher discriminatory power between brands than traditional GC-MS methods, as demonstrated by the comparative visual analysis of mass spectra, score plots and chemometric models generated in this study. Chemometric models that were trained with all five brands produced testing error rates as low as 0% and no false positives in either the testing set or the blind study; this lack of false positives is crucial for forensic approaches.

While GC-MS methods are sensitive, they are time-consuming and do not offer the same level of differentiation that is generated with DART-MS analysis of gasoline. For the comparative GC-MS methods considered in this research, analyzing one sample took approximately 26 minutes, not including the sample preparation time consisting of a 14-hour extraction and other preparative methods. Conversely, 12 gasoline samples were analyzed using DART-MS in approximately 6 minutes. No unique characteristic patterns were illustrated in the gas chromatograms, or the summed ion data generated from the GC-MS results, based on brand. However, each of the five brands: Shell, Sunoco, Irving, Cumberland Farms and Gulf had detectable CASs when analyzed via DART-MS. Additionally, the GC-MS data produced fewer ion peaks in total and no significant higher-mass ions past a range of approximately 200 amu when the summed ion data was generated. The DART-MS spectra included high-mass ions for each of the individual brands with m/z peaks as high as approximately 998, which could represent various additives. Although some of the gasoline “set 2” samples such as Irving, Sunoco and

Gulf, were highly similar through visual analysis of mass spectra derived by DART-MS, the subsequent chemometric analyses utilizing PCA and the Analyze IQ Lab software SAV algorithm were still able to generate score plots with clusters formed based on brand and models that could differentiate these brands with average testing error rates for all models falling under 6%.

Other factors demonstrated to affect the mass spectra of the five gasoline brands include the season in which the samples were purchased, the container the samples were stored in and the age of samples, all factors in addition to brand that could produce sample classifications. The tendency of gasoline mass spectra to change due to gasoline sample aging effects has been observed in other studies of gasoline analysis by DART-MS[21,39], indicating that it could be a common occurrence with gasoline; this phenomenon is also relevant to forensic case-work because samples are not always collected or analyzed immediately. While the seasonal changes and apparent aging of gasoline initially seemed to be confounding and complicating variables in this research, if a larger size of gasoline samples were to be collected encompassing all of these variables, more robust models could be trained that could offer more extensive classification by chemometric analysis.

The hypothesis that DART-MS analysis methods are capable of differentiating gasoline brands was supported by this research. This study also demonstrated that the careful optimization of DART-MS parameters is crucial to the analysis of gasoline. This approach to gasoline analysis appears to offer a greater discriminatory power based on brand as well as a rapid analysis time than other methods currently utilized in common

practice. As noted in another study using this analytical method to evaluate gasoline[40], DART-MS analysis detected many ions that remain unidentified. Although the exact nature of these chemical species observed in the mass spectra is not well elucidated, these gasoline brands do produce 'fingerprint'[40] mass spectra that can be subsequently used to identify them[40]. With additional method development more sensitive analytical methods could be further developed. The CASs produced by DART-MS and the subsequent chemometric analysis, which can be supplemented with gasoline samples of additional variables, could offer a highly discriminating and efficient classification method for gasoline samples.

5. FUTURE DIRECTIONS

This research has demonstrated that DART-MS analysis can generate chemical attribute signatures for various gasoline brands, leaving several subject areas that could be further investigated to develop a more rapid and efficient gasoline analysis method in a forensic setting. Firstly, the scope of the study could be expanded with enhanced recovery method development efforts including modified headspace concentration approaches of fire debris as well as the solvent extraction of various materials spiked with gasoline, all subsequently followed by DART-MS analysis. Activated charcoal strips sufficiently collect analytes of interest in gasoline sample analysis performed with headspace concentration methods. Because other studies suggest that higher desorption temperatures are needed to adequately desorb analytes from the activated charcoal strips, investigations could be performed by spotting both neat and diluted gasoline samples directly onto the activated charcoal strips followed by DART-MS analysis. If the

expected CASs are observed, a series of heated headspace experimentations could be carried out to determine if an optimal extraction temperature and time could be developed for an enhanced recovery of ignitable liquids from fire debris samples that are subsequently analyzed by DART-MS. Because dilutions in pentane were able to generate spectra that were distinguishable both by visual mass spectral analysis and by chemometric analysis, solvent extraction methods could also be analyzed. Solvent extraction recovery could be evaluated by spiking various materials encountered in a fire scene investigation, such as carpet, with neat gasoline and then burning the sample. Afterwards, the fire debris samples could be analyzed in accordance with ASTM solvent extraction guidelines before analysis utilizing DART-MS. Additionally, although Carbopack™ X was demonstrated to not be a suitable sorbent for the collection of target gasoline analytes, other sorbents could be analyzed.

Next, the models built using chemometric analysis software could then be supplemented with a greater sample size. This approach would include collecting samples from all five of the same gasoline brands, but from different physical service stations in the state of Massachusetts, which would determine if the gasoline from each company in a small geographic location shares the same CAS. Afterwards, samples from the same brands could be collected from other states. This sampling process could be performed with gasoline samples from both seasons. These samples could be analyzed utilizing DART-MS in the same manner as the samples analyzed in this research. This evaluation would better explain the variations in the gasoline samples. Samples of other gasoline brands could also be included as well as samples stored in various types of

containers. Additionally, these samples could be used to build more robust models that would be able to better classify more unknown gasoline samples. Lastly, though polymeric repeating units were identified in the DART-MS spectra, further identification of the higher-mass ions could be carried out to identify the components of the gasoline and further elucidate their structures. Tandem MS-MS and analysis of the samples on instrumentation capable of higher mass resolution could be employed for such identifications.

6. LIST OF JOURNAL ABBREVIATIONS

Anal. Chem.	Analytical Chemistry
Environ. Health Perspect.	Environmental Health Perspectives
Forensic Sci. Int.	Forensic Science International
J. Am. Acad. Psychiatry Law	Journal of the American Academy of Psychiatry and the Law Online
J. Am. Soc. Mass Spectrom.	Journal of American Society of Mass Spectrometry
J. Forensic Sci.	Journal of Forensic Sciences
J. Proteome Res.	Journal of Proteome Research
Ind. Eng. Chem. Res.	Industrial & Engineering Chemistry Research
Surf. Interface Anal.	Surface and Interface Analysis
Rapid Commun. Mass Spectrom.	Rapid Communications in Mass Spectrometry
TFRS	Topical Fire Report Series

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8. CURRICULUM VITAE

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